ELSEVIER

Precambrian Research 85 (1997) 173–199

Precambrian Research

Palaeomagnetic study of Neoproterozoic glacial rocks of the Yangzi Block: palaeolatitude and configuration of South China in the late Proterozoic Supercontinent

Zhang Qi Rui [1], J.D.A. Piper [2,*]

[1] *Institute of Geology, Academia Sinica, Beijing 100029, China*

[2] *Geomagnetism Laboratory, Department of Earth Sciences, University of Liverpool, Liverpool, L69 7ZE, UK*

Received 19 September 1996; accepted 19 June 1997

Abstract

Regional glaciation is recorded at two stratigraphic levels in the Neoproterozoic Sinian succession of the Yangzi (South China) Block. A palaeomagnetic study is reported here of the upper sequence (~730–670 Ma) which contains red beds. An 'A1' component ($D/I=291/-57°$, $\alpha_{95}=5.0°$, Palaeopole 151.2°E, 0.2°N) is resolved at high unblocking temperatures and resides in specular haematite. A study of contemporaneous deformation structures suggests that this component was acquired at an early stage of lithification and yields a representative palaeolatitude (37°) for the glaciation. Lower blocking temperature magnetizations referred to as 'A2' ($D/I=297/-19°$, $\alpha_{95}=7.6°$, Palaeopole 171.8°E, 19.2°S) and 'A3' ($D/I=319/-1°$, $\alpha_{95}=3.2°$, Palaeopole 169.1°E, 41.8°S) are later overprints correlating with probable Late Precambrian or Cambrian palaeofields. In other regions, the 'A' magnetizations were found to be partially or completely overprinted by a Recent field and/or a 'B' remanence ($D/I=4/13°$); the latter correlates with a remagnetization widely recorded elsewhere in South China and is attributed to the Mesozoic Indosinian Orogeny.

An ~90° apparent polar wander (APW) swathe is identified from the Yangzi Block between ~730–670 Ma and 540 Ma. It is not yet clear whether poles from older Sinian rocks record earlier magnetizations or were overprinted during this APW motion. Comparison of the Sinian–Cambrian palaeomagnetic record from South China with the contemporaneous record from Australia implies that the former block was located immediately to the north-west of Australia during these times. This reconstruction is compatible with Early Palaeozoic biogeographic data. Continuity of the ~900 Ma Cathaysian–Yangzi suture with the ~1000 Ma Albany–Fraser belt in Australia suggests an extension of this reconstruction back to the earlier part of Neoproterozoic times. A proposed location of South China adjacent to western North America (Li et al., 1995) does not conform with the palaeomagnetic data. No support is also found for the Rodinia reconstruction during the critical 750–550 Ma interval required by the stratigraphic correlation from which the reconstruction was originally derived (Jefferson, 1980). We conclude that the large-scale three-fold sedimentary cycle, which formed the basis for this reconstruction, resulted from global eustatic changes and cannot be used to infer close spatial proximity. © 1997 Elsevier Science B.V.

Keywords: Neoproterozoic; glaciation; palaeomagnetism; Yangzi Block; South China

* Corresponding author.

0301-9268/97/$17.00 © 1997 Elsevier Science B.V. All rights reserved.
PII S0301-9268(97)00031-4

1. Introduction

The last few years have seen a heightened interest in the Neoproterozoic Earth. This has come, in part, from recognition of the important and enigmatic changes in the biosphere during these times (Knoll and Walter, 1992; Dalziel et al., 1994). This interest has also prompted a re-evaluation of Neoproterozoic glaciation because glacial rocks of this age are often a part of distinctive facies associations; the diamictites frequently occur interbedded with rocks more often assigned to low latitudes such as carbonates, red beds and phosphorites (Schemerhorn, 1974; Hambrey and Harland, 1981; Harland, 1983; Fairchild et al., 1994).

Whether or not lowland ice sheets actually occupied latitudes much lower than Permo-Carboniferous and Quaternary ice has, however, been the subject of much recent debate (e.g. Meert and Van der Voo, 1994; Williams et al., 1995; Young, 1995). This debate has a bearing on such questions as the obliquity of the Earth during Neoproterozoic times (e.g. Williams, 1993). A recent palaeomagnetic study of ~600 Ma Marinoan periglacial rocks in South Australia has established unequivocally that they were formed close to the ancient equator as defined by the axial geocentric dipole model (Schmidt and Williams, 1995). It is therefore important to extend modern palaeomagnetic studies to Neoproterozoic glacial rocks elsewhere to test whether or not a low latitude origin is more generally applicable to these glacial deposits (Kirschvink, 1992).

The other major palaeogeographic question relating to Neoproterozoic times concerns the configuration of the continents. Whilst it is widely accepted that most, or all, of the continental crust was aggregated together into a single large supercontinent during these times (Piper, 1982; Bond et al., 1984; Piper, 1987), the models proposed are diverse. Most influential has been the SWEAT reconstruction of Moores (1991), Hoffman (1991) and Dalziel (1991) which places Eastern Australia against North America. This reconstruction has been assigned the name 'Rodinia' following McMenamin and McMenamin (1990), although it differs significantly from the configuration proposed by these latter authors.

The existence of a Rodinian supercontinent has been widely debated in recent years: whilst it proposes some regional correlations between mobile belts and stratigraphic successions, it is associated with several serious difficulties. In particular, only two Neoproterozoic poles from Australia can be invoked to support it and younger data require a separation of Australia and North America after ~750 Ma (Powell et al., 1993). There is no clear geological expression for a breakup event and the stratigraphic basis for correlating Eastern Australia and Western North America is based primarily on post-750 Ma successions (Jefferson, 1980; Eisbacher, 1985). If these correlations imply a close spatial configuration in Late Neoproterozoic times, major differential movements of continental crust are then required to achieve the Gondwana reconstruction by latest Proterozoic–early Cambrian times. Thus to the advocates of Rodinia, the evidence for the SWEAT correlation has outweighed contrary evidence that Gondwana was already essentially an integral continental unit prior to the Pan Africa mobile events (e.g. Kroner, 1977; Piper, 1987).

A specific aspect of the SWEAT reconstruction is the proposition that South China (the Yangzi Block and Cathaysian Fold Belt) was also situated close to the western margin of North America and broke away from other continents at ~700 Ma at about the time of the Sinian glaciations (Li et al., 1995). Well-constrained palaeomagnetic poles are required to test this model. The present investigation of Neoproterozoic glacial rocks in South China has aimed to do this and to determine the palaeolatitude of diamictite formation.

2. Geological framework

The major tectonic divisions comprising China are shown in Fig. 1. The South China Block (SCB) was sutured to North China (the Huabei or Sino-Korean Block) along the Qinling Orogenic Belt during Upper Palaeozoic to Mesozoic times (Zhao et al., 1996). A close biogeographic link between the SCB and the northern margin of Gondwana,

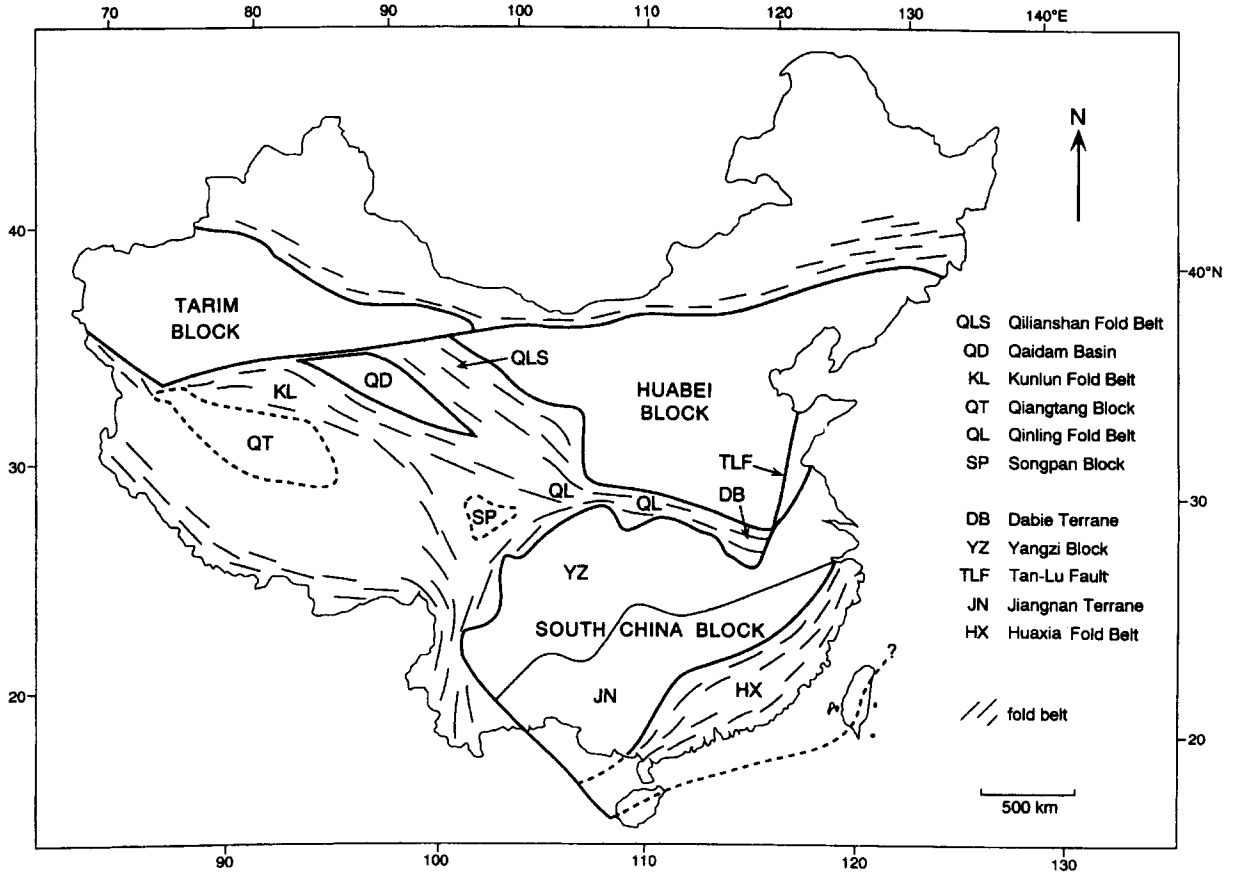


Fig. 1. Outline tectonic map of China showing the major Precambrian blocks and the Late Neoproterozoic and Palaeozoic terranes.

specifically Australia, appears to have existed in Lower Palaeozoic times prior to this suturing event (Burrett and Richardson, 1980; Long and Burrett, 1989).

Neoproterozoic glacial deposits are widely recorded within each of the three major palaeocontinental units. These are the Yangzi, Huabei (also referred to as the North China or Sino-Korean) and Tarim Blocks (Fig. 1). The best documented examples occur in the Yangzi and Huabei Blocks and the rock successions including glacial deposits from the Yangzi Block are the focus of the present investigation. A feature of the diamictites which emphasizes the peculiar, and possibly unique, nature of Neoproterozoic glacial deposits, is the development of a red pigment. The widespread presence of haematite in these rocks makes them suitable for palaeomagnetic study and they comprise the bulk of the samples investigated here.

The Neoproterozoic geology of the SCB and distribution of sampling locations are shown in Fig. 2. Consolidation of the metamorphic basement terranes comprising the Yangzi Block is generally thought to have been completed by the Jinning Orogeny which occurred between 1100 and 800 Ma (Hu et al., 1991; Liu and Zhang, 1991): recent assessments of the geological evidence suggest that the Huaxia Fold Belt to the south-east was amalgamated with the Yangzi Block prior to 800 Ma (Chen et al., 1995) along the Jiangnan Terrane which is sited between them (Fig. 2).

During mid-Sinian times, a NW to SE transition from terrestrial to marine environments was present across the southeast part of the Yangzi Block

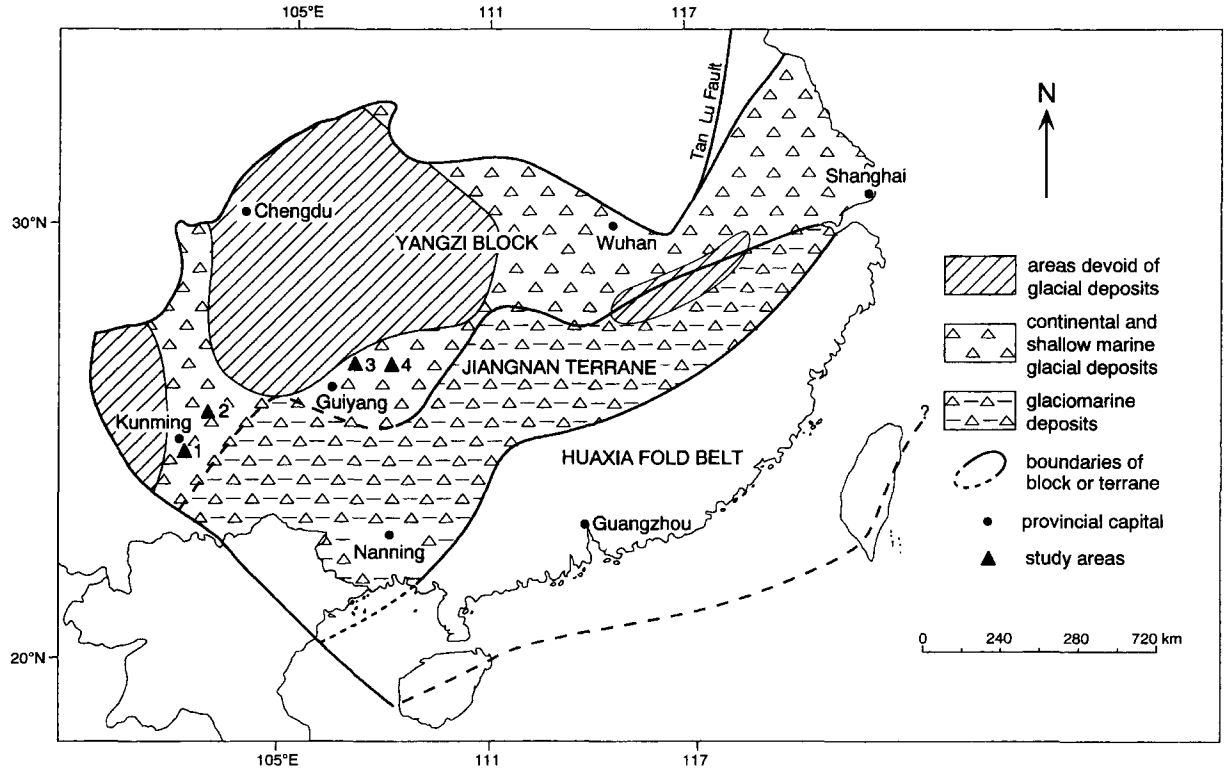


Fig. 2. Distribution of Neoproterozoic glacial rocks in South China showing the location of the four areas investigated here. See Fig. 1 for the regional location.

and glacial red beds were deposited within the terrestrial portion. Only one glacial level (always the upper one) is preserved here and the thickness of this horizon is usually less than 200 m. In contrast, contemporaneous glacial deposits are more fully developed within the Jiangnan Terrane to the south east where two glacial levels plus an interglacial horizon are commonly seen. This latter succession attains a thickness of several thousands of meters; unfortunately none of these rocks are reddened and they are probably unsuitable for palaeomagnetic study. Further to the south east, outcrops of Neoproterozoic rocks occur within the Huaxia Block although their glacial affinity is contentious (Yang et al., 1986).

The stratigraphy of the Sinian Supergroup in South China is summarized in Table 1. The Lower Sinian succession within the western part of the Jiangnan Terrane comprises the Banxi Group; this is actually more complete that the type section of Lower Sinian exposed along the Yangzi Gorges which comprises the Liantuo Formation (Liu, 1991; Liu et al., 1993; Tang, 1993 and Table 1). The upper limit of the Sinian Supergroup is dated 540 Ma by Compston et al. (1992) and a lower limit of 935 Ma has been proposed by Gan et al. (1993). Owing to a paucity of suitable rock types, dating of the glacial periods is difficult. The older estimate of ~730 Ma is based on a U–Pb single-grain zircon date of 736 ± 2 Ma from keratophyre tuff near the top of the Banxi Group (Gan et al., 1993) which agrees with a zircon U–Pb date of 748 ± 12 Ma on the upper part of the Liantou Formation (Ma et al., 1980). The younger limit of 680–670 Ma is based on a Rb–Sr whole rock isochron from black carbonaceous shales in the lower part of the Doushantuo Formation and is probably least reliable. Hu et al. (1991) summarize the existing results and propose a younger limit of 670 ± 10 Ma for Sinian glaciation. The older age

Table 1
The Sinian Chronostratigraphic Succession in the Yangzi Block

Stratigraphic age	Rock unit	
Lower Cambrian	Qiongzhusi Formation	540 Ma
Upper Sinian	Dengying Formation	(Ediacaran Fauna)
	Doushantuo Formation	
Middle Sinian	Nantuo Formation	670–680 Ma
	Datangpo Formation	(Upper glacial level)
	Fulu Formation	(Interglacial facies)
	Chang'an Formation	(Lower glacial level)
Lower Sinian	Liantuo Formation	730±5 Ma
	(unconformity)	
	Lengjiaxi Group	935 Ma

limit on the upper glacial level investigated here (the Nantuo Formation) is not yet known, but is believed to lie between 730 and 670 Ma.

The glacial and related deposits were sampled at four locations. The first (comprising 91 cores from sites 1–6 and indicated as region 1 in Fig. 2) is in Malong County, Yunnan Province, near the southern border of the county (Fig. 2). The Nantuo Formation (see stratigraphic columns in Fig. 3) is composed of two members; the lower member is mainly purple diamictite and the upper is red shale and fine-grained sandstone. The second study area is sited north of the county town of Chengjiang in Yunnan Province. Seven sites (7–13, 117 cores, region 2 in Fig. 2) comprise a top to bottom section through the Chengjiang Formation commencing just below the diamictite and passing downwards into coarse purple sandstones. The Chengjiang Formation is generally considered to be coeval with the Liantuo Formation (Liu and Zhang, 1991); on this assumption it predates the entire glacial episode (Table 1).

The third area is the Kaiyang Phosphorous Mine (KYPM, location 3 in Fig. 2). The stratigraphic location of the succession here is still controversial: it was formerly considered to represent the pre-glacial Liantuo Formation, but rare dropstones have been found, Liu (1991) and one of us (ZQR) believe that it was deposited at the time of the glaciation and may therefore correlate with the Nantuo Formation. Since the lithology is different from the type Nantuo and comprises purple to red siltstones and shales with little diamictite, the name 'Maluping Formation' is adopted. The outcrop in the KYPM area is a north–south trending anticline and one section was sampled on each limb: sites 17 and 18 on the east limb are stratigraphically lower than sites 19 and 20 from the west limb located near the top of the unit (Fig. 3).

The fourth area is located in Wong'an County (location 4 in Fig. 2) where the Nantuo and Doushantuo Formations are condensed to about 30 m. The lithology of the Nantuo Formation here contrasts with the other sampled regions: a lower member comprises 15 m of grey–green coarse feldspathic sandstones and an upper member comprises 5 m of purple red diamictite (sampled at site 21, 9 cores). At the top of the dimictite is a dolomite assigned to the Doushantuo Formation which incorporates a bed of purple shale (Fig. 3) just 2 metres above the lower boundary (site 22, 18 cores).

3. Field and laboratory methods

A widespread problem to be addressed by palaeomagnetic sampling in southern China is the profound degree of sub-tropical weathering. Fresh exposures are hard to locate and mostly found in quarries and newer road cuttings; the bulk of the present sites were located in such exposures. A further problem constraining the sampling strategy is the recognition of extensive Mesozoic remagnetization within the South China Block (e.g. Wang et al., 1993; Wang and Van der Voo, 1993). This was probably caused by fluid activity during the

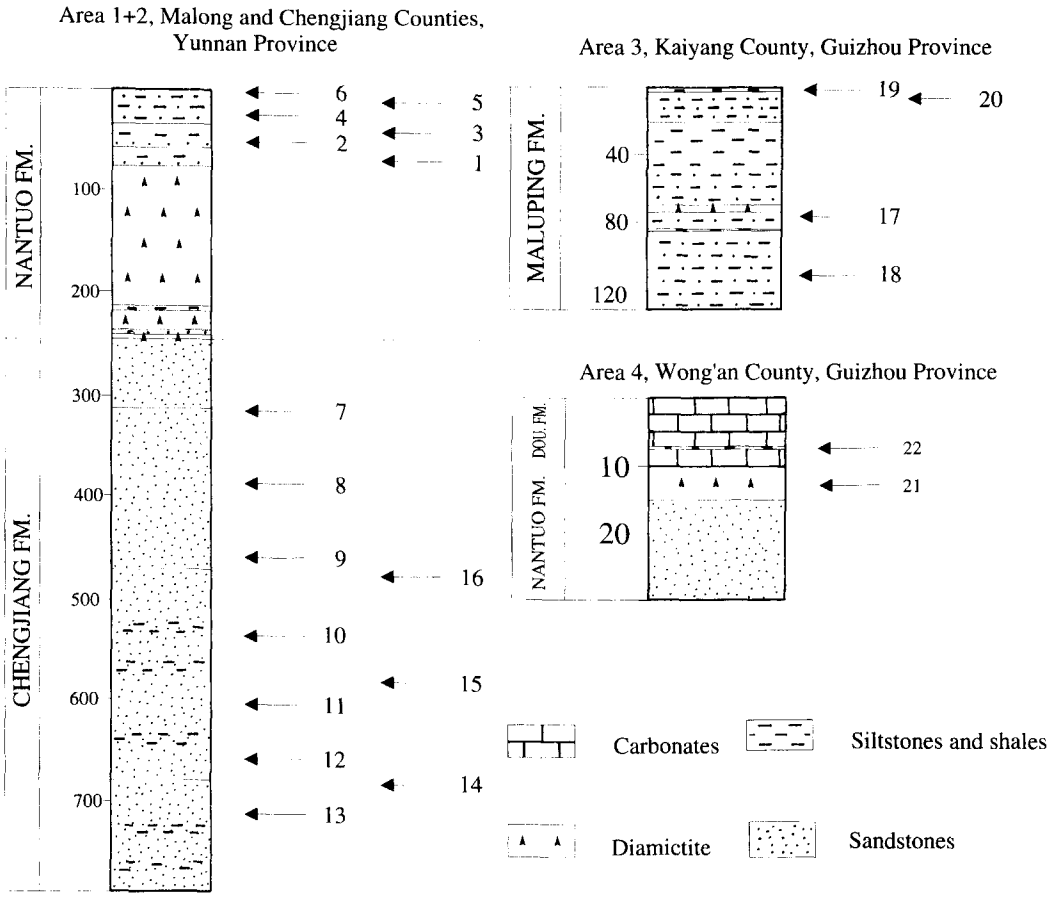


Fig. 3. Stratigraphic sections of study areas 1–4 in the Yangzi Block (Fig. 2) showing levels of the sampling sites and dominant lithologies. Numbers to the left of the lithology column are thicknesses in metres from top to bottom.

Indosinian Orogeny which accompanied collision and suturing of the South China Block into the continental collage of eastern Asia (c.f. McCabe and Elmore, 1989). Studies of orogenic remagnetization elsewhere have observed a relationship to advancing fluid fronts imparting secondary remanence at various stages of folding. Whilst remanence may be determined as pre-, syn- or post-deformational by reference to folded structures, it is ultimately part of the same event (Setiabudidaya et al., 1994). The best possibility for recovering a pre-orogenic remanence is afforded by the least-strained rocks. In the present example, these comprise little-deformed platformal successions within the interior of the block. Because the conventional palaeomagnetic fold test is not then effective, penecontemporaneous deformational structures have been used to assess the age of magnetization.

Samples were collected as short cores with portable drills and oriented by Sun and magnetic compasses prior to extraction; these were subsequently cut into 2.3 cm length cylinders for palaeomagnetic processing. Magnetizations were measured by SQUID magnetometer and all cores were subjected to progressive thermal demagnetization using Magnetic Measurements (MM2D) demagnetizers. The thermal demagnetization proceeded in steps of 50 or 100°C to 300°C and subsequently in diminishing steps of 30, 25 or 10°C to the Curie points of the magnetic carriers or until magnetization behaviour ceased to be systematic. The entire procedure was conducted within a large set of

Rubens coils with the ambient magnetic field continuously nullified to minimize secondary remanence acquisition during treatment. Components constituting the natural remanent magnetization (NRM) were resolved by examination of orthogonal projections of the demagnetization results. They were then isolated and their directions calculated by principal component analysis. Unidirectional and three-dimensional isothermal remanent magnetization (IRM) studies of selected samples were also made to evaluate carriers of magnetization.

4. Rock magnetic investigation

The complex magnetization histories typical of many red sediments are a consequence of processes occuring during prolonged lithification and diagenesis. Various phases of haematite are the usual products of these processes and may include detrital specularite, pigmentary films and intergranular authigenic grains. Owing to their high coercivities and Curie point close to 680°C, these phases are not well discriminated by most rock magnetic techniques. Isothermal remanent magnetization (IRM) study is the most sensitive method for detecting trace concentrations of haematite (Robertson and France, 1994) and has been applied here to highlight types and relative amounts in these rocks.

Selected samples were magnetized using a pulse magnetizer. The applied fields were increased in approximately logarithmic steps (2, 5, 10, 15, 20, 25, 30, 40, 50, 80, 100, 120, 160, 200, 250, 300 mT···) until a saturation IRM (SIRM) was achieved or the upper limit of the field in the magnetizer (2500 mT.) was reached. Subsequently samples were incrementally magnetized by the same process in the opposing direction until the IRM was reduced to zero. At this point, the applied field value defines the coercivity of the remanence (H_{cr}). The peak field in the pulse demagnetizer is unable to fully saturate haematite although it is adequate to identify the essential behaviour of this mineral. The acquisition of IRM is best described by a combination of curves of cumulative log Gaussian (CLG) functions. An acquisition curve can be described by a combination of the individual CLG functions when multiple phases are present (Robertson and France, 1994). Curves are observed to be a combination of at least two cumulative functions in the Sinian red sediments. This is illustrated in Fig. 4 using an adaptation of the method first described by Eyre (1994) in which the gradient of the acquisition curve is plotted against log of the applied field. Three parameters describe each CLG function and are listed in Table 2 for the studied samples. The SIRM of the phase is the height of the log Gaussian curve; $B_{1/2}$ is the applied field at which 50% of the SIRM of the particular mineral phase is acquired, and DP is a statistic describing the dispersion of apparent coercivities of the mineral phase. Parameters $B_{1/2}$ and DP are intrinsic properties related to the type of mineral phase and independent of concentration (Robertson and France, 1994), whereas SIRM is extrinsic and volume-dependent, relating to the dilution factor of the magnetisable mineral phase within the sample.

Normalized IRM acquisition curves indicate that all samples have similar mineral phase compositions. The dominant phase has $\log(B_{1/2})$ values in the range 2.5 and 3.0, typical of haematite. Coercivities of this phase would be compatible with grains of detrital specularite or authigenic origin. There is a second phase with $\log B_{1/2}$ values between 2.00 and 2.35. This is consistent with a fine-grained (pseudo single domain–single domain) ferrimagnetic mineral probably pigmentary haematite. Large SIRMs are observed in red sandstones of the Nantuo Formation overlying the diamictite in Yunnan where the sum of SIRM values is in the range 0.001–0.006 (Table 2). Elsewhere in the Yangzi Block the redbed formations have values of less than 0.002. These observations typically accord with remanent intensities retained during thermal demagnetization (last column in Table 2) with the first three samples having much higher intensities than the remainder.

Relative contributions of high and low coercivity components to the remanence can be evaluated by thermal demagnetization of an orthogonal three component IRM (Lowrie, 1990). IRMs of 2.5, 0.4 and 0.1 T are applied in that order along mutually

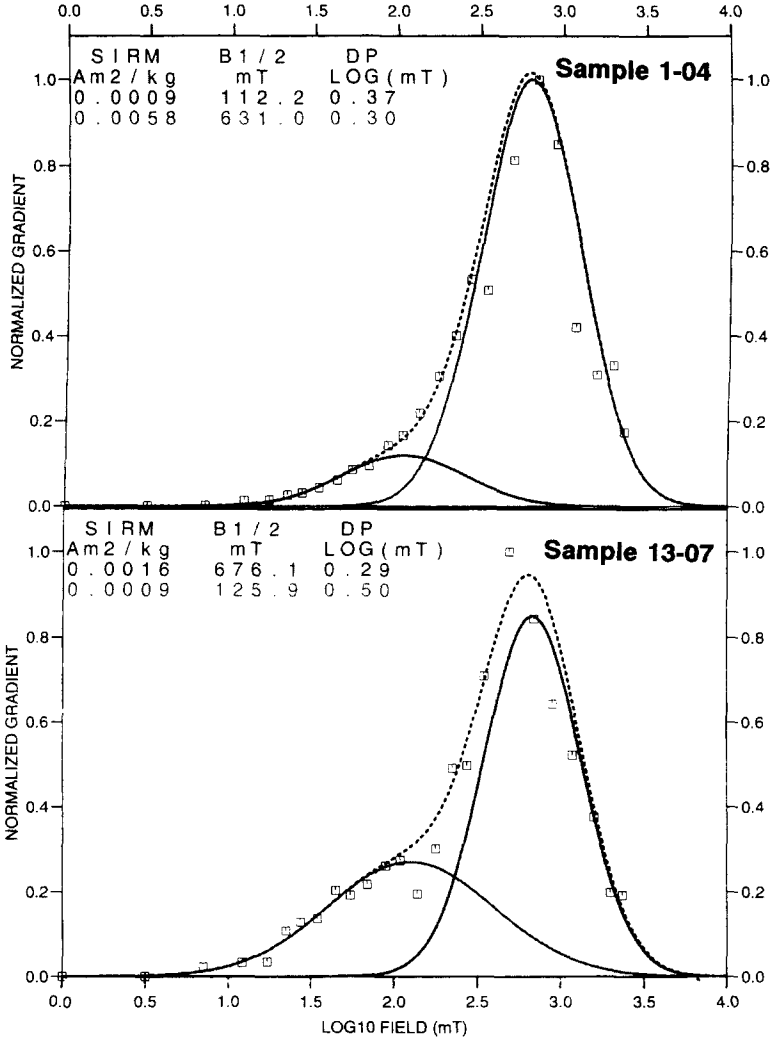


Fig. 4. Gaussian functions fitted to normalized gradient of IRM in two examples of rocks from this study.

Table 2
Results of IRM analysis

Sample number	First mineral phase			Second mineral phase			H_{cr}	J^{a}_{400}
	SIRM	$\log(B_{1/2})$	*DP*	SIRM	$\log(B_{1/2})$	*DP*	(mT)	
1–04	0.0009	2.05	0.37	0.0058	2.80	0.30	569.91	0.528
4–20	0.0018	2.15	0.45	0.0021	2.73	0.35	359.08	1.801
6–11	0.0010	2.30	0.45	0.0037	2.78	0.30	567.19	0.424
13–07	0.0009	2.10	0.50	0.0016	2.83	0.29	492.30	0.124
17–25	0.0005	2.10	0.50	0.0014	2.82	0.26	565.93	0.036
20–07	0.0003	2.00	0.37	0.0015	2.74	0.27	512.55	0.019

[a] J_{400} is the intensity of magnetization ($\times 10^{-5}$ Am^2kg^{-1}) after thermal demagnetization to 400°C.

orthogonal directions and referred to as hard, medium and soft components respectively. These rocks are dominated by medium and high blocking temperature components with narrow unblocking temperature (T_B) spectra. Maximum unblocking temperatures are ~650°C confirming the dominant presence of haematite. The only suggestion of another phase is the change in gradient of medium IRM curves between 200 and 300°C, The source of this feature cannot be positively identified; possible causes are fine grained haematite near the superparamagnetic–single domain transition or the transformation of goethite (formed from near surface weathering) into haematite (Fig. 5).

There is no substantial evidence for a magnetizing phase other than haematite in any of these rocks. We conclude that the remanence carrier is haematite with little or no magnetite. A.f. demagnetization is therefore unlikely to be effective and only thermal demagnetization was carried out. The complex paragenesis of haematite also identifies the need for caution when relating remanence to geological events; thus, haematite formation can relate to deposition, early (~100–500 years) dewatering (Tauxe and Badgley, 1984), or to much later diagenesis (Larson and Walker, 1982).

5. Palaeomagnetic results

Up to three components are recognized in orthogonal projections of the demagnetization data. Linear segments tend to be discretely-defined without prominent overlap. An early remanence comprises the bulk of the unblocking temperature spectrum in sandstones of the Purple Red Shale Member overlying the Nantuo Diamictite in Malong County (Fig. 6). Northerly positive magnetizations attributable to acquisition in a Recent field ('R' components) are removed by 400–500°C (samples 1–05, 1–10, 2–04 and 6–03 in Fig. 6). The mean of these components (Table 3) is 9° shallower than the present field ($I=+44.5°$) suggesting possible contamination with a northerly shallow ('B') remanence identified at other locations in south China (see below) and interpreted as a Mesozoic overprint. Higher T_B components are mostly convergent and either directed NW with shallow negative inclination (1–05, 2–04 in Fig. 6) or have more westerly declinations (usually) and steeper negative inclinations (1–10, 4–11, 5–04, 6–03 and 6–08 in Fig. 6). In some component structures, an intermediate T_B component with a NW-direction is unblocked to recover a high T_B $W-$ component (e.g. 2–04, 6–03 and 6–08 in Fig. 6); the inclination of the latter is typically steeper than the former. A contoured distribution of all components belonging to this population highlights a concentration of NW intermediate to steep negative directions separated from a population of components with shallow negative inclinations and more dispersed declinations (Fig. 7). A paucity of inclinations at ~30° separates these groups. Hence in this analysis, we distinguish two populations 'A1' and 'A2' by $I>30°$ and $I<30°$, respectively. In Malong County, all A1 and A2 components are of one polarity.

Components belonging to both populations can have high and narrow unblocking T_B spectra concentrated just below the Curie point of haematite. However, whereas A1 is always unblocked above 620°C, A2 may be unblocked over a wide range of temperatures and is evidently less thermally-stable. Furthermore some samples show evidence for removal of A2 before isolation of A1 (e.g. 1–10, 6–03 and 6–08 in Fig. 6). At site 2, only A2 is present, whilst at other sites, A1 and A2 are present in different cores from the same site and less commonly in the same sample (Table 3).

Tectonic tilts are less than 15° everywhere and are barely perceptible at sites 3–5, where they cannot confidently be resolved from depositional dips. Accordingly, although tilt-adjusted directions are used to compute the overall mean in Table 2, no meaningful fold test is possible. However, site 3 is within a sandstone unit with prominent syn-sedimentary deformation; 10 of 12 cores yield good definition of the A1 component with demagnetization between 520 and 680°C. Inspection shows that components from the northerly dipping limbs of the syn-sedimentary folds are slightly displaced from components resolved from cores on the southerly dipping limbs (Fig. 8). A fold test may therefore be performed on these data at sample level. Values in situ ($\xi_2=5.353$) and fully tilt-adjusted ($\xi_2=8.268$) are both larger than the criti-

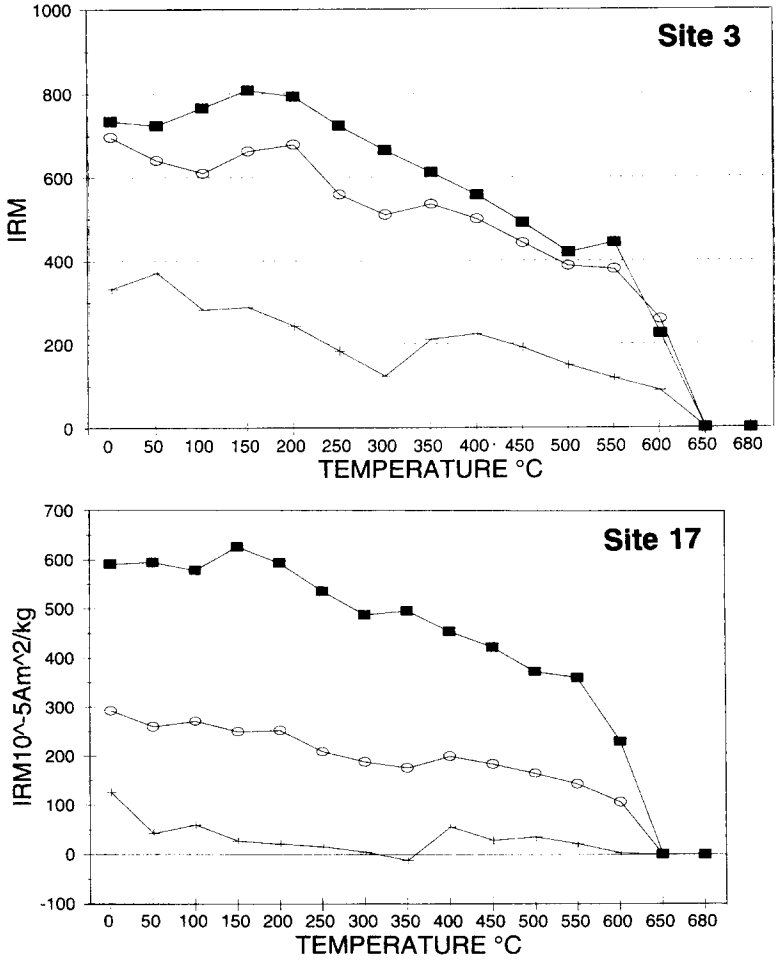


Fig. 5. Examples of thermal demagnetization of three component IRMs. The squares are hard components, the ovals are medium components and the crosses are soft components.

cal value of the test statistic at the 95 and 99% confidence levels ($N=10$, McFadden, 1990). With 24% unfolding however, ξ_2 achieves a minimum value of 0.192; this is within 2% of the value at which maximum precision is also attained (Fig. 8) and indicates that A1 was fixed prior to lithification before some, but not all, of this syn-sedimentary slumping was complete. A post-depositional detrital (PDRM) origin is therefore suggested for this remanence; uniformly narrow and high blocking temperature spectra in these samples (e.g. 3–10 in Fig. 6) would be compatible with residence in detrital specularite grains.

Red beds of the Chengjiang Formation underlie the Nantuo diamictite with disconformity. They yield a remanence record predominantly of secondary origin. This is apparent as a relatively large component in all samples and was acquired in the Recent field (see, for example, 10–09, 11–08, 12–06 and 13–11 in Fig. 9). The mean of these ('R') components is close to the mean field direction ($D/I=0/43°$) at this location (Fig. 10). In some cores (e.g. 12–07 in Fig. 9), this is the only component present. In other samples, a small residual component of northerly direction and shallow inclination is resolved exclusively in the haematite spectrum (11–08, 13–15, 13–21 and 14–09 in Fig. 9). This is recognized in approximately one

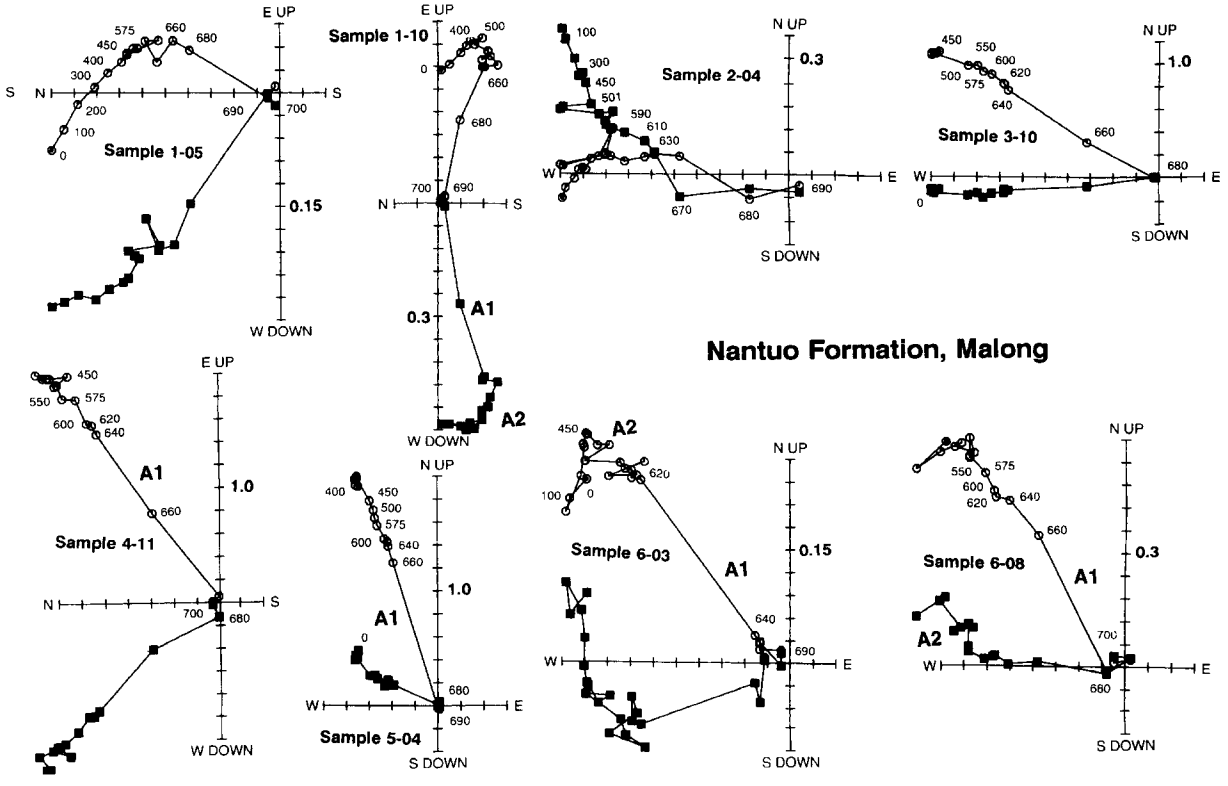


Fig. 6. Examples of thermal demagnetization behaviour of samples from the Nantuo Formation in Malong County. Magnetization vectors are projected onto the horizontal plane as solid squares and the vertical plane as open circles. Demagnetizing temperatures are in °C and intensities are $\times 10^{-5}$ A m^{2}kg^{-1}. Directions are shown in situ. Examples of component definitions (A1, A2, etc) are indicated in this and subsequent figures.

third of the collection and is the second most important component at this location; it is referred to as the 'B' population.

Evidence for survival of A1 and A2 in this formation is also apparent in a similar number of samples to B, but is always resolved over a much narrower unblocking temperature spectrum and mostly within 50° of the Curie point of haematite (e.g. 8–02, 8–18 and 12–06 in Fig. 9). In a few cores where three component structures can be positively resolved, this is a residual high blocking temperature feature (sample 13–11). The presence of some antiparallel examples of A1 in this formation identify this palaeofield axis as dipolar [Fig. 10(b)]. In Table 4, the A1 and A2 populations are subdivided on the basis of inclination (> or <30° respectively) as at locality 1.

Little of the remaining Chengjiang Formation exposure proved fresh enough for study. A single site (16), located 40 km to the north, does not conform to the synthesis derived from the remaining sites near Chengjiang. A NW directed component of shallow inclination is resolved here over a distributed blocking temperature spectrum (16–06 in Fig. 9) and is possibly an early Mesozoic overprint although it has a declination ~20° more westerly than typical fold-associated remagnetizations in South China (Wang and Van der Voo, 1993). Six large residual components isolated above 620°C have consistent NW positive direction (A4) yielding a tilt-adjusted palaeopole (74°E, 60°N) of possible late Sinian age (Section 6).

Angles of tectonic tilt in the Chengjiang outcrop are less than 30° and mostly directed to the W–NW quadrant. Thus, fold tests are not diagnostic (Table 4), although, since directions contrast with

Table 3
Site and group mean palaeomagnetic results, Nantuo Formation, Malong County, South China (Area 1, 103°E, 25° 20'N)

Site number	Comp.	D	I	D	I	N	R	α_{95}	k
		(In situ)		(Tilt adjusted)					
1	A1	277.4	−46.6	261.1	−48.0	4	3.79	24.9	14.6
	A2	291.3	−13.1	287.3	−19.6	14	12.94	11.8	12.3
2	A2	285.9	−12.3	283.8	−18.0	7	6.30	21.9	8.6
3	A1	297.7	−49.2	299.3	−55.5	10	9.49	11.8	17.6
	A2	329.4	−6.2	330.0	−11.0	2	1.87		
4	A1	296.7	−49.0	299.7	−50.6	15	14.10	10.0	15.6
	A2	293.7	−14.9	294.4	−16.7	7	6.40	20.3	9.8
5	A1	305.0	−66.2	306.3	−69.6	12	11.13	12.7	12.7
	A2	337.1	−5.0	337.3	−7.6	2	1.94		
6	A1	296.4	−50.2	280.0	−56.6	14	13.43	8.5	22.7
	A2	310.6	−11.1	308.0	−21.1	10	9.36	13.3	14.1
Mean, A1 components									
		295.4	−51.9	290.7	-56.6	59	55.12	5.0	14.9
Palaeomagnetic pole: 151.2°E, 0.2°N (dp/dm=5.4/7.5°)									
Mean, A2 components									
		298.6	−12.7	296.7	−18.8	43	38.45	7.6	9.2
Palaeomagnetic pole: 171.8°E, 19.2°S (dp/dm=4.2/8.0°)									
Mean, R components									
		356.7	35.7			27	25.03	7.9	13.2
Pole: 312.2°E 83.7°N (dp/dm=5.3/9.2°)									

D and I are the mean declination (E) and inclination (+ down, − up) derived from N samples. R is the magnitude of the resultant vector and k is the Fisher precision parameter [$=(N-1)/(N-R)$]. α_{95} is the radius of the cone of 95% confidence about the mean direction. Dp and dm are the radii of the oval of 95% confidence about the pole position derived from the mean direction in the colatitude direction and at right angles to it respectively.

Mesozoic and later palaeofields, tilt-adjustment for the probable Indosinian deformation is favoured. As a consequence of weak deformation, groupings of A1 and A2 directions are barely affected by tilt adjustment whilst the 'B' component is optimized at 0% unfolding (ξ_1=2.29 in situ and 9.62 tilt adjusted) providing no reason to reject the hypothesis that this remanence is a post folding overprint.

A pre-Mesozoic remanence is less apparent in the Maluping Formation of the Kaiyang region. Distributed unblocking temperature spectra are dominated by northerly positive components with a mean close to the present field inclination [I= 45.5°, Fig. 11(c)]. A smaller number of 'B' components with more dispersed declinations is also present. Although some of the Recent field components are not convergent [17–17, 18–24 and 20–01 in Fig. 11(a)], recovery of component directions in the small residual remanence is only rarely successful due to magnetomineralogical changes within the clay fraction of these fine to medium grained sandstones. Both normal and reversed examples of the A magnetizations are suggested (18–08 and 18–24 in Fig. 11), but trajectories are poorly defined. A few components of both polarities attributable to A2 are poorly resolved at the highest levels of successful demagnetization (samples 17–24 and 18–23 in Fig. 11, but are too few to merit interpretation (Table 5).

Although rare dropstones were identified in this formation, deformed siltstones sampled from beneath a dropstone yielded no examples of 'A' remanence to permit a soft sediment test for age of remanence. The sampled region is gently deformed into a low anticline with sites 17 and 18 dipping 18–27° to the east and sites 19 and 20 at

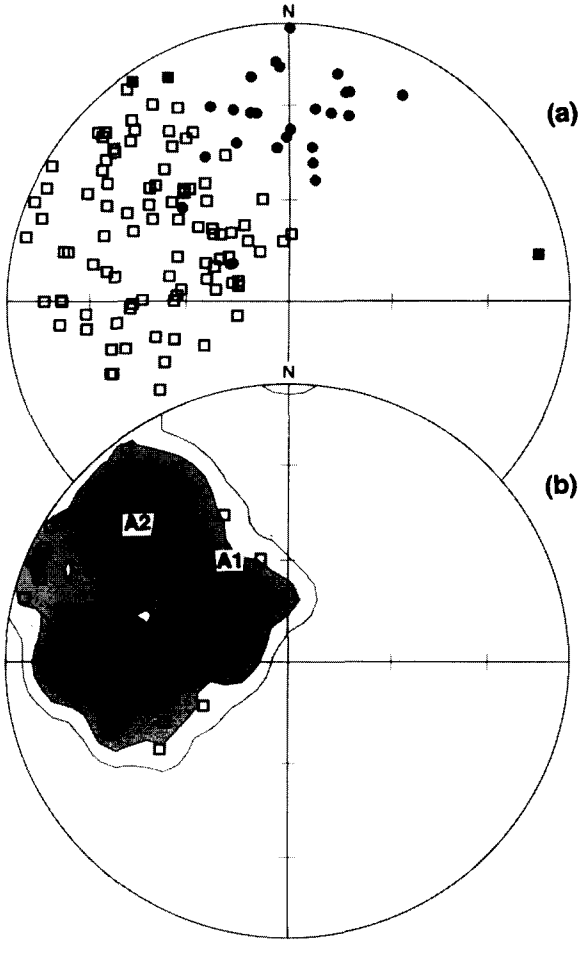


Fig. 7. (a) Equal-area projection showing the distribution of in situ component directions resolved from the Nantuo Formation in Malong County. Closed symbols are lower hemisphere and open symbols are upper hemisphere plots. The northerly positive directions (in situ, closed circles) are interpreted as Recent or older Phanerozoic overprints whilst the squares (tilt adjusted) are assigned to the 'A' population. (b) Contoured distribution of components components belonging to the A1 and A2 populations (in situ).

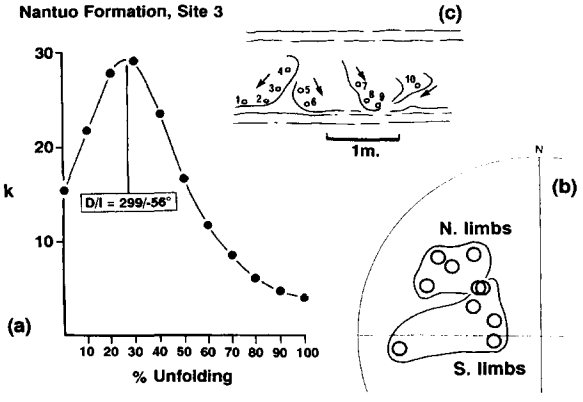


Fig. 8. (a) Precision parameter, k, as a function of partial unfolding of a syn-sedimentary slump in sandstones at site 3; (b) in situ directions of A1 components resolved from 10 of 12 cores at this site with directions derived from the N dipping limbs distinguished from those on S dipping limbs; and (c) field sketch of this site showing distribution of sampled cores. The section is oriented south to north from left to right; cores 11 and 12 to the right of this section did not yield A1 ChRMs.

21–23° to the west. Unfortunately, because the B direction is close to the axis of the fold, adjustment for tilt is inconclusive.

Red beds of the Nantuo and Douchantuo Formation in Wong'an County either have no Recent field overprint [22–13 in Fig. 12(a)] or a minor one removed below 300°C (21–08, 22–14). The ChRM has NW declination and positive inclination; this is close to convergent in all cores. This NW positive in situ direction has not been recognized elsewhere in the study (with the possible exception of three cores from the Malung area, see Fig. 7) and is referred to 'A3'. Both sites dip to the west at angles of 40–50° and were probably deformed at the time of the Indosinian Orogeny in Mesozoic times. Since A3 in situ is directed to the west of the Mesozoic and younger palaeofield migration path for South China, we suspect that it is pre-folding in age. The initial direction of this remanence is therefore NW with near zero inclination [Fig. 12(b) and Table 6]. A small number of cores with near-zero in situ inclinations (Table 6) merit no detailed comment and might include unresolved contributions from an A1 magnetization.

6. Interpretation

6.1. Magnetizations derived from this study

The A1 component is recognized in rock formations both above (Malong, Area 1) and below (Chengjiang, Area 2) the younger of the two Sinian diamictite levels in the Yangzi Block. It is the most

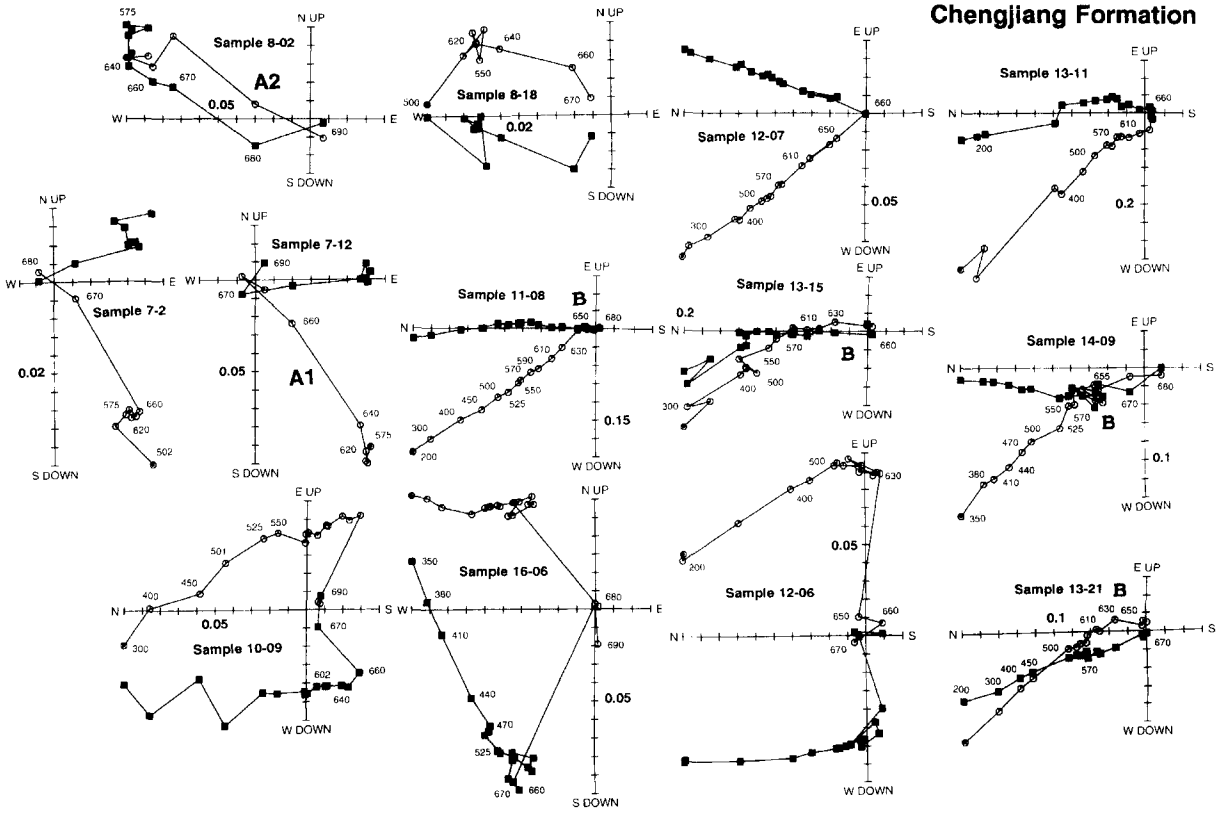


Fig. 9. Demagnetization behaviour of samples from the Chengjiang Formation. Directions are in situ and symbols are as for Fig. 6.

thermally stable remanence and is always recovered above the Curie point of magnetite in the highest part of the haematite spectrum. In the Malong region, it appears to be a primary remanence dating from lithification and implies that the diamictite was emplaced at a palaeolatitude of $37 \pm 9°$. Whilst this is typical of the latitudes of Phanerozoic redbed formation, no regional ice-sheet formation in such low latitudes has been recognized during Phanerozoic times at sea level (Frakes, 1979). It contrasts however, with the equatorial palaeolatitude established for the Late Neoproterozoic Marinoan glaciation in Australia (Schmidt and Williams, 1995; Williams et al., 1995) but is compatible with a range of estimates for Neoproterozoic regional glaciation at intermediate latitudes above 25° (Meert and Van der Voo, 1994).

Li et al. (1991) interpret low inclinations in sediments above and below Sinian glacial deposits from the Tarim Block to indicate deposition in much lower palaeolatitudes ($\sim 8°$) than our result from South China. There are several possible explanations for this discrepancy including: (1) the Tarim results refer to the lower of the two Sinian glacial levels; (2) Tarim and South China were widely separated within the Late Proterozoic Supercontinent; and (3) the Tarim examples are later remagnetizations. In consideration of the latter possibility, we note that the results come from sites close to the northern tectonized margin of the Tarim Block and the pole derived from the 'in situ' direction (248°E, 40°N) is close to poles from the mid-Triassic overprint in the South China Block (Wang and Van der Voo, 1993). Unfortunately, the paucity of mid-Triassic to mid-

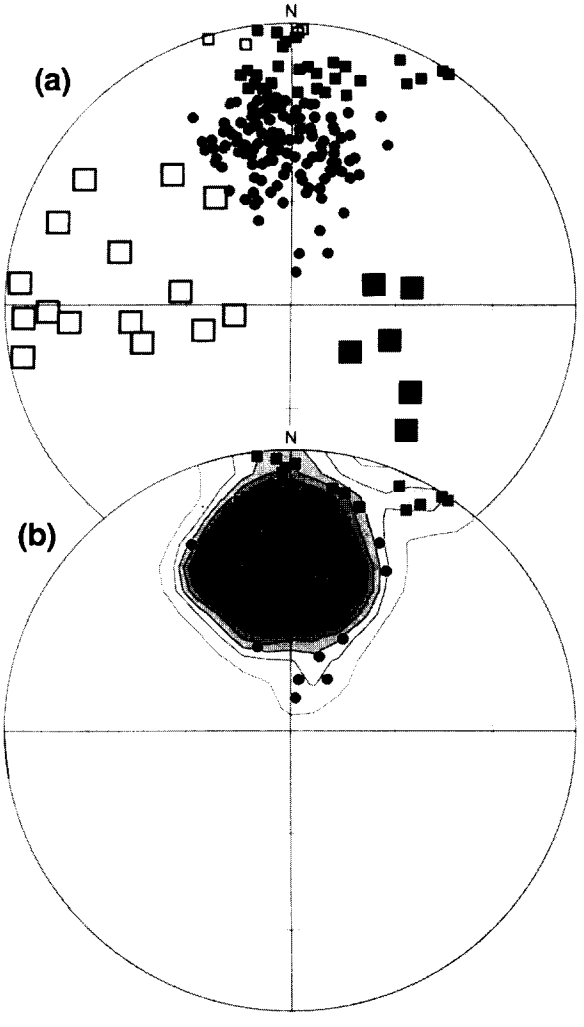


Fig. 10. (a) ChRM directions from the Chengjiang Formation. Solid symbols are lower hemisphere plots and open symbols are upper hemisphere plots. Circles are components ($I>30°$) interpreted to be overprints acquired in the present field; small squares are 'B' components interpreted to be Mesozoic overprints and large squares are high blocking temperature components assigned to A1 and A2 components. (b) Contoured plot of Present Field and 'B' components.

Cretaceous poles from Tarim does not at present permit this possibility to be evaluated (Zhao et al., 1996).

The A2 and A3 magnetizations are less thermally-stable than A1 and presumably later overprints. They do not, however, correlate with the known Mesozoic and later APW from the South China Block (Fig. 13). Hence, they appear to predate Mesozoic deformation and pole positions are interpreted following adjustment for tilt. They are possibly Late Neoproterozoic or Cambrian in age. However, a later Palaeozoic age cannot be excluded because Opdyke et al. (1987) report a remanence ($D/I=86/12°$) from Silurian siltstones of the Yangzi paraplatform, which is fairly close to the reversed A2 direction.

The B magnetizations are readily correlated with early Mesozoic overprints widely observed in South China (Wang and Van der Voo, 1993) and attributed to remagnetization during the Indosinian Orogeny. In Triassic sediments of Guizhou Province, similar pre-folding and post-folding magnetizations are recognized in Middle Triassic sediments (Wang et al., 1993), which differ only marginally from the B magnetizations of this study.

6.2. *Neoproterozoic-Cambrian poles from the South China Block*

In their recent assessment of Phanerozoic palaeomagnetic data from China, Zhao et al. (1996) interpret four Late Precambrian–Early Cambrian poles from South China. Three of these are from the Meishucun Formation, a carbonate and phosphorite succession of Late Precambrian to Early Cambrian age in Yunnan Province (Lin et al., 1985; Fang et al., 1988; Liu et al., 1988). The poles contrast with each other and provisionally identify large APW of this age although none of these studies can demonstrate the age of magnetization with certainty. A fourth study of early Cambrian carbonates of Zhejiang, Hubei and Yunnan Provinces identified dual polarity and passed a fold test (Lin and Fuller, 1990). These data are complemented by a study of carbonates spanning the Late Precambrian–Cambrian boundary in Guizhou Province (Wang et al., 1995) where an 'A' remanence considered to be nearest to the age of the rocks bears no close similarity to any of the other data noted above (Table 7). A 'B' remanence in these formations corresponds to the Mesozoic B overprint also recognized in the present study but a third ('C') remanence may be Cambrian or, since it also plots fairly close to

Table 4
Group mean palaeomagnetic results, Chengjiang Formation, Chengjiang, South China (Area 2, 103° 30'E, 25° 40'N)

Site numbers	Component	D	I	D	I	N	R	α_{95}	k
		(In situ)		(Tilt adjusted)					
7	B	16.9	16.3	12.0	11.6	6	5.86	11.6	34.6
9	B	355.4	5.4	357.2	3.8	7	6.83	10.4	34.4
10	B	351.3	20.7	340.8	17.5	5	4.99	3.3	524.8
13	B	10.7	11.0	11.0	−5.9	7	6.70	13.8	20.0
14	B	9.6	13.7	25.7	14.2	4	3.91	16.4	32.3
16	–	333.9	3.7	331.9	41.0	11	10.81	6.4	51.6
16	A4	338.3	28.5	338.4	66.5	6	5.86	11.5	34.8
Palaeomagnetic pole: 74.1°E, 60.5°N, dp/dm=15.6/18.9°									
Mean A1 components									
(In situ)		283.9	-44.4			14	13.15	9.6	15.3
(Tilt adjusted)				287.8	−56.0	14	13.35	9.1	15.7
Palaeomagnetic pole: 153.4°E, 2.2°N, dp/dm=9.4/13.1°									
Mean A2 components									
(In situ)		276.4	0.2			7	6.37	17.2	9.5
(Tilt adjusted)				276.4	−11.7	7	6.66	15.1	16.9
Palaeomagnetic pole: 185.4°E, 3.2°S, dp/dm=7.8/15.3°									
Mean B components									
(In situ)		4.1	13.0			33	31.71	5.1	24.8
Palaeomagnetic pole: 270.6°E, 71.5°N, dp/dm=2.7/5.2°									

Only site mean groupings for $N>3$ are listed. Symbols are as for Table 3.

Silurian and Devonian poles from South China (Fig. 12), could be a younger Palaeozoic overprint. The A2 overprint from this study is similar and has a comparable interpretation.

In Table 7, we group the Neoproterozoic to Lower Cambrian palaeomagnetic results from the South China Block according to stratigraphic location of the source rocks within the Sinian succession. As noted above, only the higher glacial level is recorded within the Yangzi Block. This is separated by hiatus from rocks below and all formations beneath the higher diamictite (including the Chengjiang) are regarded as pre-glacial. In addition to the results noted above, only two poles from post-glacial Sinian rocks include thermally-demagnetized data and could be relevant to these times (poles PI and LA, Zhu and He, 1987). Furthermore, the three poles PI, XN and LA in Fig. 13, which plot remote from the remaining data, come from southern Anhui, which is probably a part of the allochthonous Huaiyu Terrane (Zhang Huimin, personal communication). Four of the five reported results from the Nantuo and the coeval Silikou Formations (Liu and Liu, 1965; Li and Liu, 1979, Liu et al., 1993; Zhang et al., 1993; see Table 7) appear to record only Mesozoic overprinting, whilst the fifth result from a single site (pole N4) has shallower inclination than the A1 remanence of this study and is possibly an A2 magnetization contaminated with a younger overprint (Fig. 13).

The results from older Neoproterozoic rocks in South China listed in Table 7 are incompletely reported and difficult to assess; they are based mostly on results from single sites and tests to constrain the age of remanence are unavailable. Furthermore, results are derived from a variety of facies including slates in which the carriers of remanence are unclear. However, these poles are removed from the Carboniferous and later APW path from South China (Fig. 13). They may therefore record a Late Sinian (post Nantuo) pole

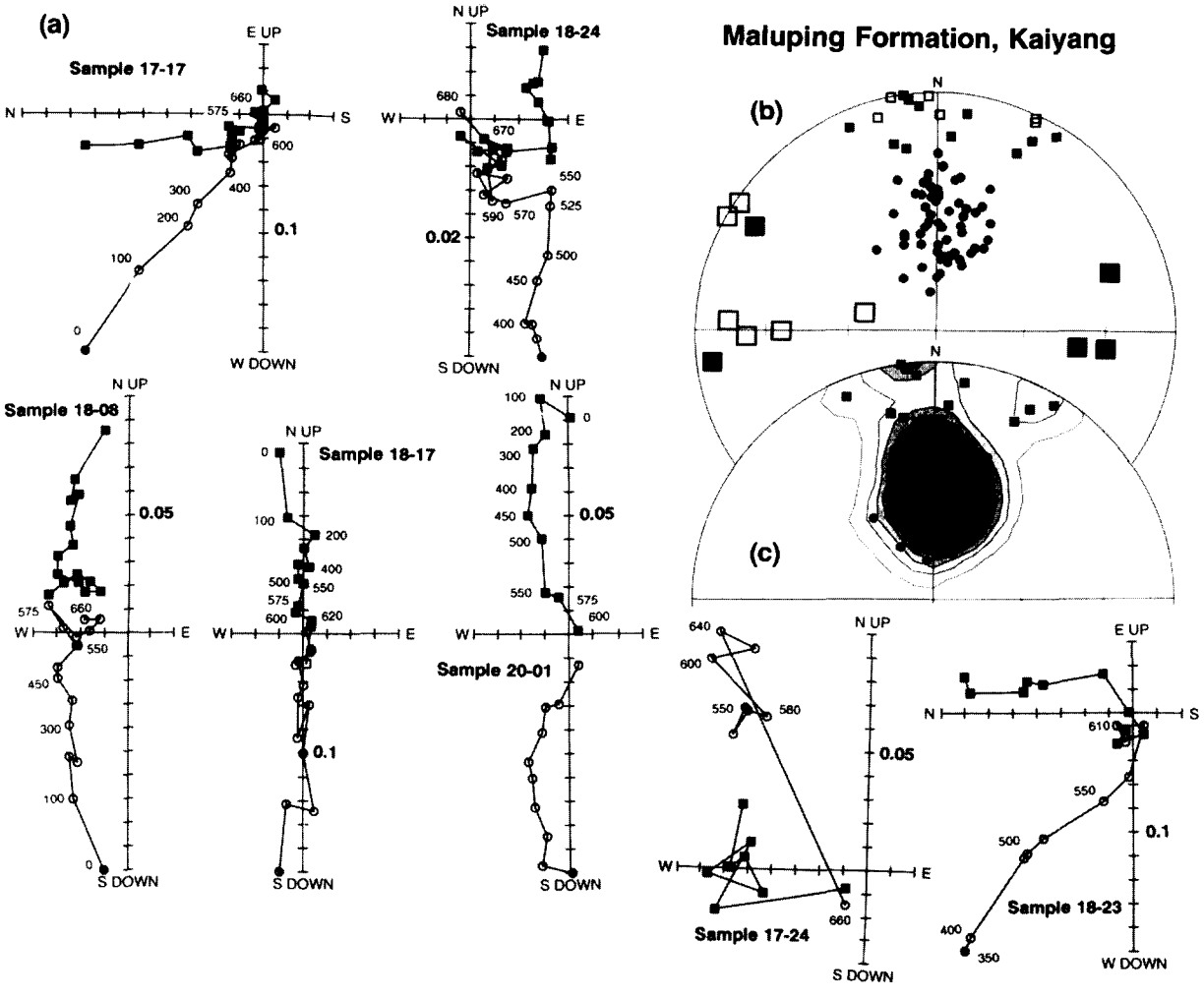


Fig. 11. (a) Orthogonal projections illustrating typical demagnetization behaviours of samples from the Maluping Formation in the Kaiyang area. Directions are in situ and symbols are as for Fig. 6. (b) Directions of ChRMs derived from this area. Small circles are interpreted as overprints in the Recent field, small squares are 'B' magnetizations and the large squares are residual high blocking temperature components assigned to A2. (c) Contoured plot of Recent and 'B' magnetizations from this location.

position. Alternatively, the recognition that most of them are similar to the Nantuo A2 (pole N2) and Gechongwu C (pole GC) overprints derived from the two most detailed investigations of the Neoproterozoic in South China suggests that they could record a regional overprinting event. Whether this was Late Sinian or Siluro-Devonian in age is not clear at present (Fig. 13). The location of the poles on the path connecting the Nantuo A1 pole (N1, 730–670 Ma) and Gechongwu A pole (GA, Late Precambrian–Cambrian boundary) supports the former possibility.

Subsequent poles from Latest Precambrian to Lower Cambrian formations (poles TI, Y1, Y2, Y3 and ZA in Fig. 13), imply, at face value, that large APW occurred during Lower Cambrian times. This conclusion has been drawn from Lower Cambrian results on several other continental blocks (McElhinny et al., 1974; Piper, 1987; Park, 1992). However, in the case of the SCB, the data

Table 5
Site and group mean palaeomagnetic results, Maluping Formation, Kaiyang, South China (Area 3, 106° 40'E, 27°N)

Site number	Components	D	I	D	I	N	R	α_{95}	k
		(In situ)		(Tilt adjusted)					
17	B	18.9	13.6	21.6	4.9	4	3.89	17.7	27.4
18	B	4.3	7.6	7.3	4.8	6	5.78	15.4	19.8
19	B	350.2	22.1	346.1	10.8	3	2.94	22.3	31.7
20	B	355.4	6.1	3.9	10.7	4	3.88	18.7	25.0
Mean, B components									
(In situ)		3.3	10.3			17	16.18	8.3	19.5
(Tilt adjustment)				2.8	6.2	17	16.04	9.0	16.7
Palaeomagnetic pole: 278.0°E, 68.0°N (dp/dm=4.2/8.4°)									

Symbols are as for Table 3.

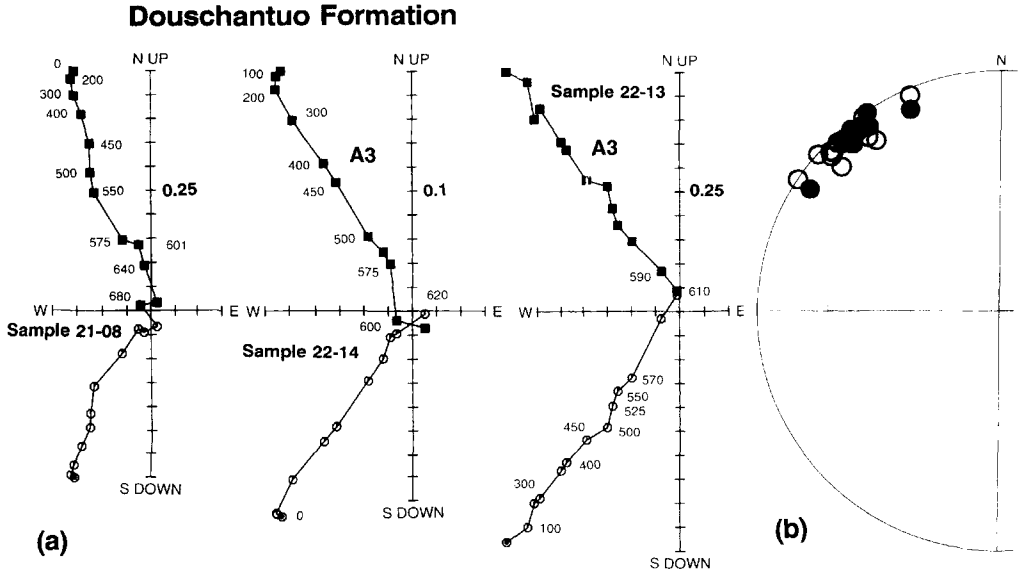


Fig. 12. (a) Orthogonal projections illustrating demagnetization behaviours of samples from the Doushantuo Formation. Directions are shown in situ and the symbols are as for Fig. 6. (b) Tilt adjusted ChRM directions assigned to the A3 population.

on which this inference hinges cannot positively be discriminated from later APW; more results constrained by field tests are therefore required to assess Cambrian APW of this block.

6.3. South China in Gondwana

Palaeomagnetic correlations between the constituent blocks of Gondwana are hampered by poor age control and heavy reliance on older data in probable need of revision (see Meert et al., 1995 for recent reviews). The most comprehensive studies of the Late Neoproterozoic and early Cambrian have been undertaken in Australia where results come from the Adelaide System and the Amadeus Basin in SE and central Australia respectively (e.g. Klootwijk, 1980). An essential agreement between these two Late

Table 6
Site and group mean palaeomagnetic results, Nantuo and Doushantuo Formations, Wong'an, South China (Area 4, 107° 18'E, 27°N)

Site number	Component	D	I	D	I	N	R	α_{95}	k
		(In situ)		(Tilt adjusted)					
21	A3	332.2	37.6	322.8	4.5	9	8.86	7.0	55.6
22	A1/3	322.9	4.3	332.3	−32.0	6	5.89	9.8	47.3
	A3	326.1	35.2	316.8	−4.5	16	15.88	3.4	120.3
Mean, A1/3 components[a]									
		325.9	6.0	333.6	−28.6	8	7.84	8.6	42.8
Palaeomagnetic pole: 141.8°E, 40.6°S (dp/dm=5.2/9.5°)									
Mean, A3 components									
		328.2	36.1	318.9	−1.3	25	24.71	3.2	81.8
Palaeomagnetic pole: 169.1°E, 41.8°S (dp/dm=1.8/3.6°)									

[a]Includes two components from site 21. Symbols are as for Table 3.

Precambrian–Cambrian data sets precludes major differential movements between the intra-Australian regions, and the collective poles plot along a small circle arc spanning some 90°. Although this trend may embrace more complex APW movements, the Cambrian results lie at the SE part of the swathe and suggest a general clockwise APW motion (Fig. 14).

The trend of Sinian poles from South China can be matched to these contemporaneous Australian data with the SCB sited immediately to the north west of Australia. The reconstruction shown in Fig. 14 employs a rotation of +68.5° about an Euler pole at 11°N, 146.5°E and places the SCB near the northern margin of Gondwana consistent with a biogeographic correlation (Burrett et al., 1990).

There is scope for small regional rotation of South China within this constraint: the Sinian poles plotting at the NW extremity of the combined track are magnetizations for which the possibility of Mesozoic remagnetization has been noted. Most of the Sinian poles correlating with the Australian Cambrian results are from late Neoproterozoic rocks and the Cambrian data from the SCB are more scattered. Furthermore, SE China was characterized by intermediate to deep-water facies during Cambrian times (McKerrow et al., 1992) and probably sited at an outward facing margin of this part of Gondwana. A partial anticlockwise rotation of the SCB to achieve this would be acceptable provided that overlap of the Neoproterozoic portions of the Sinian and Adelaide–Amadeus tracks is preserved. The location of these great circle tracks implies that the South China–Australia land mass was sited in low latitudes (<30°) throughout the Late Neoproterozoic to Middle Cambrian interval (McKerrow et al., 1992). The entire track as shown in Fig. 14 could therefore be the expression of anticlockwise rotation of Australia–South China whilst this unit remained sited in low latitudes.

The suture zone between the Cathaysia and Yangzi Blocks is recognized as a Mesoproterozoic orogen from ages of 1034 and 935 Ma on ophiolite suites (Chen et al., 1995). Post-collisional granites are mostly in the range 950–900 Ma with some as young as 800 Ma (Chen et al., 1995; Zhou et al., 1993). This belt is brought into continuity with the Albany–Fraser and Musgrave mobile zones of Australia of comparable age suggesting that the reconstruction shown in Fig. 14 dates back to at least ~950 Ma.

6.4. South China and Rodinia

Intercontinental correlations between the Late Mesoproterozoic orogenic/mobile belts have formed a key element in the Late Precambrian

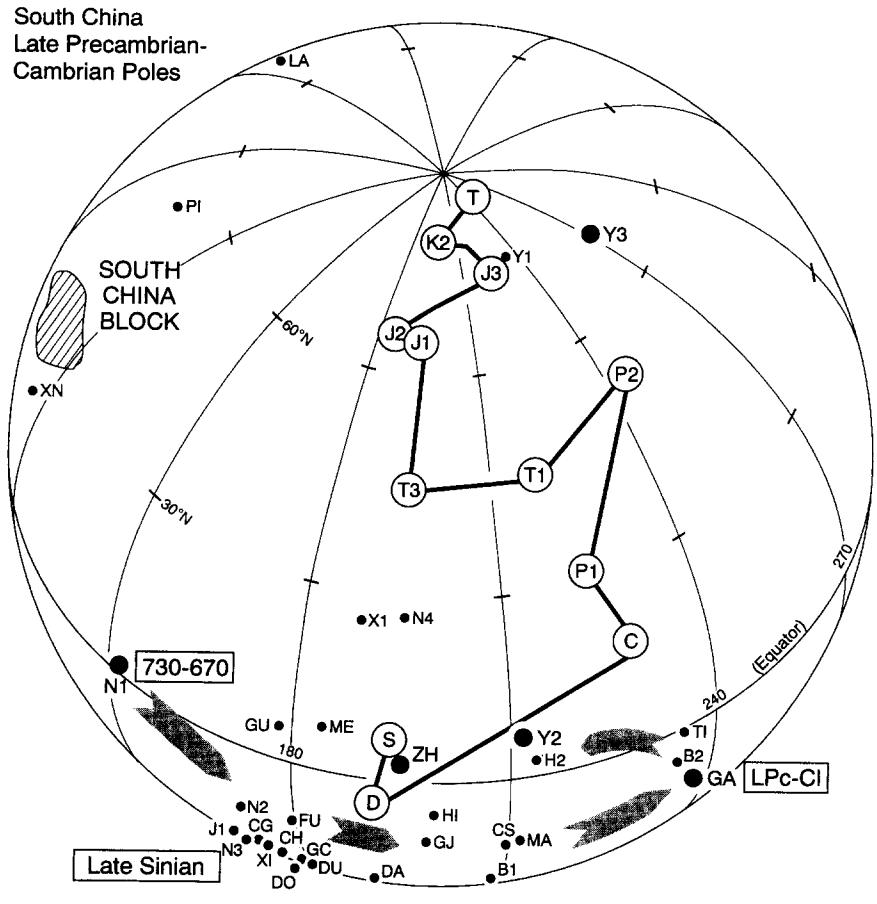


Fig. 13. Neoproterozoic–Cambrian apparent polar wander (APW) path from the Yangzi Block. The pole positions are referenced according the the prefixes in Table 7. The Phanerozoic APW path from the South China Block is from mean poles listed by Zhao et al. (1996).

supercontinent reconstructions (Piper, 1987; Hoffman, 1991) and several alternatives are possible. Li et al. (1995) have proposed that the SCB formed a crustal element in the Rodinia reconstruction sited between eastern Australia and Laurentia. Mean Neoproterozoic poles for Laurentia after Powell et al. (1993) and Meert et al. (1995) are summarized in Table 8. When these poles are rotated into the South China configuration proposed by Li et al. (1995) and shown in Fig. 15, no correlation is found between them and the South China–Adelaide–Amadeus poles (Fig. 15). We are therefore forced to conclude that this reconstruction has no palaeomagnetic validity. A similar conclusion applies when the Laurentian poles are rotated into the Australian reference frame: the only similarity between the Australian and Laurentian data sets is the proximity of poles from the Stuart and YB dykes from Australia, dated at ~1050 and ~730 Ma, respectively, with the Laurentian Track (c.f. Fig. 14) as noted by Powell et al. (1993).

The older age of the South China–Adelaide–Amadeus Track is unconstrained but is unlikely to exceed 700 Ma. The (probably older) YB and Nantuo A1 poles are displaced from this track (Fig. 14) and it may therefore be legitimate to compare only post ~700 Ma poles from

Table 7
Summary of Late Precambrian–Cambrian Palaeomagnetic Pole positions derived from the South China Block

Code, rock unit	Age	λ	A_{95}	Pole		Reference
				°E	°N	
(i) Late Precambrian–Lower Cambrian						
ZH Zhejiang	Cl	1.1	8.8	195	3	Lin and Fuller (1990)
TI Tianheban	Cl	25.3	20.0	235	−1	Liu et al. (1993)
GA Gechongwu A	Cl	46.0	7.3	239	−17	Wang et al. (1995)
GC Gechougwu C	Cl	0.5	11.4	179	−32	Wang et al. (1995)
Y3 Yunnan	LPc	3.7	6.6	271	69	Fang et al. (1988)
Y2 Yunnan	LPc	13.9	10.2	211	7	Lin et al. (1985)
Y1 Yunnan	LPc	15.1		227	74[b]	Liu et al. (1988)
(ii) Post glacial results from Sinian rocks						
N3 Nantuo A3		0.7	2.7	169	−42	This study
N2 Nantuo A2		9.7	6.1	172	−19	This study
Dengying[a]		9.7	11.4	223	48[b]	Zhang et al. (1983)
Doushantuo[a]		9.5	11.8	216	38[b]	Zhang et al. (1983)
PI Piyuancun		60.9	7.0	102	51	Zhu and He (1987)
LA Lantian		36.5	19.3	52	42	Zhu and He (1987)
(iii) Sinian upper glacial horizon						
N1 Nantuo A1	730–670	37.1	6.5	151	0	This study
N4 Nantuo		17.4	11.5	195	27	Zhang et al. (1993)
Nantuo[a]		14.9	6.0	216	57[b]	Liu et al. (1993)
Nantuo[a]		18.5	5.3	209	55[b]	Liu et al. (1993)
Silikou		13.3	11.9	200	31[b]	Zhang et al. (1993)
Nantuo[a]		19.3	18.7	192	24[b]	Zhang et al. (1993)
(iv) Sinian interglacial horizon						
ME Menghushan		25.5	7.9	184	9	Zhang et al. (1993)
X1 Xiangmeng[a]		21.8	17.8	188	26	Zhu and He (1987)
Xiangmeng[a]		13.2		242	70[b]	Liu et al. (1993)
(v) Sinian lower glacial horizon						
XI Xieshuihe		2.0	15.4	172	−37	Zhang et al. (1993)
CH Changtan		1.6	16.6	176	−37	Zhang et al. (1993)
J1 Jiangkou		9.3	13.0	168	−32	Zhang et al. (1993)
Jiangkou		19.7	9.1	208	60[b]	Liu et al. (1993)
CG Chang'an		4.8	12.2	172	−37	Zhang et al. (1993)
FU Fulu		10.4	21.1	359	17	Zhang et al. (1993)
GU Gucheng[a]		22.5	8.3	178	5	Zhu and He (1987)
(vi) Sinian pre-glacial rocks						
Liantuo		0.8	7.3	224	29[b]	Zhang et al. (1991)
XN Xiuning		77.8	10.6	114	18	Zhu and He (1987)
H1 Huashan	843[c]	0.9	6.4	18	10	Zhang et al. (1991)
H2 Huashan	844	8.1	9.7	34	−2	Zhang et al. (1991)
MA Machaoyuan		18.1	9.0	32	17	Zhang et al. (1993)
B1 Banxi		21.9	12.9	207	−37	Zhang et al. (1993)
CS Changshuipu	933[c]	7.0	15.9	210	−17	Zhang et al. (1993)
B2 Banxi	885—-834[d]	2.9	16.4	236	−9	Zhang et al. (1993)
GJ Gaojian	890	3.8	10.0	197	−15	Zhang et al. (1993)
DA Danzhou	837[c]	5.3	13.0	190	−31	Zhang et al. (1993)
DO Dongmen		4.8	11.1	178	−35	Zhang et al. (1993)
A4 Chengjiang, Site 16		41.5	17.3	254	−61	This Study

[a]Treatment a.f. only or not stated.
[b]Pole corresponds to Mesozoic APW path and may be overprint of this age. Age determinations are indicated. [c]U–Pb. [d]Rb–Sr.

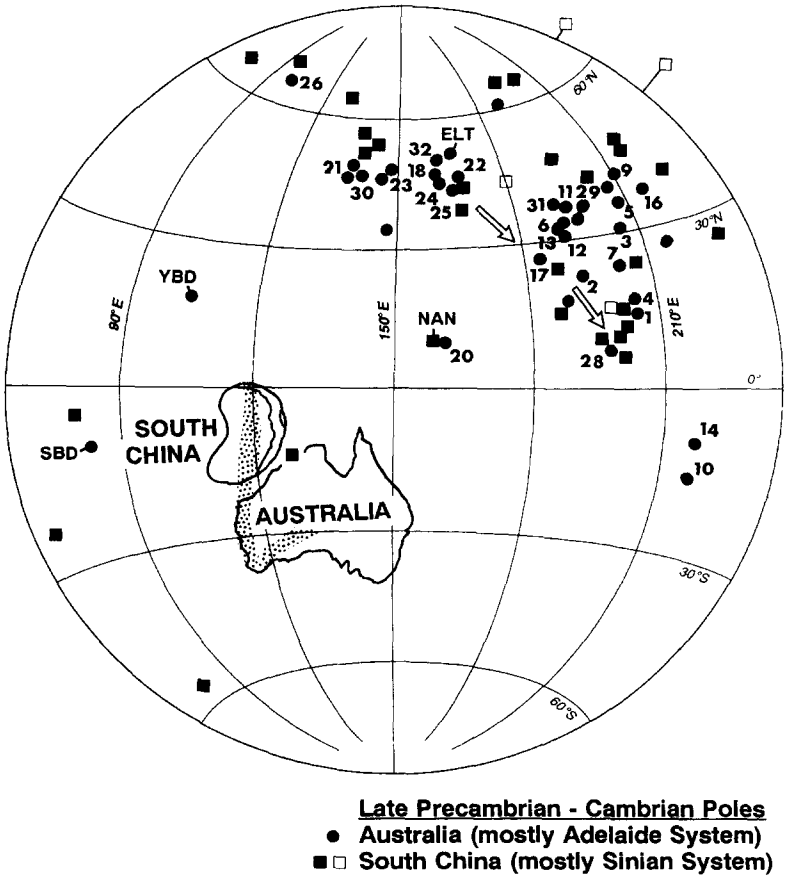


Fig. 14. Late Neoproterozoic–Cambrian palaeomagnetic poles from Australia (●) and South China (■ and □) with the latter block rotated +68.5° about a Euler pole at 11.0°N, 146.5°E. The South China results from Late Neoproterozoic (Sinian) rocks are shown as closed squares and the results from Cambrian formations are open squares (see Table 7). Identified poles are lettered: ELT, Elatina Formations varvites (Schmidt and Williams, 1995); SBD, Stuart Block Dykes (~1050 Ma); YBD, YB Dykes (730 Ma); and NAN, Nantuo Diamictite (730–670 Ma). The Australian Late Neoproterozoic to Cambrian poles defining the Adelaide–Amadeus swathe are numbered: 1, Jinduckin Formation (O); 2, Dundas Group (Cu–Ol); 3 and 4, Lake Froome Groups 1 and 2 (Cm); 5, Pantapinna Formation (Cm); 6, Areyonga Section (Cm); 7, Billy Creek Limestone (Cm); 8 and 9, Giles Formation 1 and 2 (Cl–m); 10, Shannon Formation (Cl–m); 11, Tempe Formation (Cl–m); 12, Ilcara Formation (Cl–m); 13, Deception Formation (Cl–m); 14, Hugh River Shale (Cl–m); 15, Kangaroo Island sediments (Cl); 16, Aroona Dam Sediments (Cl); 17, Hawker Group (Cl); 18, Todd River Dolomite (Cl); 19, Pound Quartzite (Lpc–C); 20, Antrim Plateau Volcanics (LPc–Cl); 21, Northamton Dykes (1040–550 Ma); 22, Elatina Formation varvite (LPc); 23, Marinoan Sandstone (LPc–C); 24, Upper Arumbera Sandstone (LPc–C); 25, Lower Arumbera Sandstone (LPc); 27, Branchina Formation (LPc); 28, Bunyeroo Formation (LPc); 29, Tapley Hill Formation (LPc); 30, Merinjina Tillite (LPc); 31, Copley Quartzite (LPc); and 32, Elatina Formation (680 Ma). Assigned ages in brackets are abbreviated: LPc, Late Precambrian; C, Cambrian; O, Ordovician; and l, m, u are Early, Middle and Late divisions respectively. The stippled region in South China is the suture between the Yangzi and Cathaysian Blocks and in Australia highlights the Frazer–Albany–Musgrave mobile zones of Late Mesoproterozoic age (see text).

Laurentia with this swathe. Even so, there is demonstrably no correlation between the two data sets and it must be concluded that either: (1) the Rodinia reconstruction of Hoffman (1991) and Dalziel (1991) had broken up by ~700 Ma (Powell et al., 1993); or (2) it has no validity for any part of Neoproterozoic times. We favour the second possibility for two reasons. Firstly, geological evidence for breakup at ~750–700 Ma is not obvious; the widespread glaciations are clearly not

Table 8
Mean Laurentian reference poles for Neoproterozoic–Cambrian times and their position in Australian and South China frames according to the Rodinia reconstruction

Age of pole (Ma)	Pole position		Pole in Australian frame		Pole in South China frame	
	°E	°N	°E	°N	°E	°N
810±40	324	11	255.5	−31.0	92.5	−64.5
780±5	318	6	263.0	−35.5	75.5	−63.0
725±15	342	−6	268.0	−9.0	165.0	−87.5
580±20	304	44	211.0	−46.0	106.5	−28.5
550±15	345	13	248.0	−11.5	145.5	−69.5
505±15	344	−3	264.5	−8.5	164.0	−85.0

Laurentian poles are rotated to the Australian frame by −132.1° about a Euler pole at 126.1°E, 28.9°N (Dalziel, 1991) and into the South China frame by a rotation of −199° about a Euler pole at 149°E, 48°N (derived from the reconstruction of Li et al., 1995).

specific indicators of breakup although the palaeogeographic changes they produce might induce glaciation after a long time lapse (c.f. Li et al., 1995; Young, 1995). Secondly, the correlations upon which the Rodinia reconstruction has been advocated refer mostly to post-750 Ma sedimentary successions in Australia and western North America (see Eisbacher, 1985, fig. 10, and Bell and Jefferson, 1987, fig. 1). This point seems to have been overlooked by some in widespread promotion of the Rodinia reconstruction during recent years. We conclude that the three large scale sedimentary cycles recognized in the Neoproterozoic successions of Australia and western North America (Jefferson, 1980; Eisbacher, 1985; Bell and Jefferson, 1987) were probably driven by global eustatic changes and do not imply close proximity of the two regions.

7. Conclusions

Although profound magnetic overprinting in South China during Mesozoic times has presented many difficulties for attempts to recover an older record from this block, we have found that unstrained rocks from the Sinian Supergroup sporadically preserve Neoproterozoic magnetizations. The character of the most thermally-stable remanence, and its apparent relationship to early post-depositional effects, indicate that deposition of diamictites in close temporal proximity took place at a palaeolatitude of ~37°. This glaciation is considered to have occurred between 730 and 670 Ma (Hu et al., 1991). Although the palaeolatitude is higher than that recorded in a later Neoproterozoic glaciation near the Neoproterozoic-Cambrian boundary (Schmidt and Williams, 1995; Piper and Zhang, 1996), it is well below the latitudes of glaciation recorded at sea level during Phanerozoic times.

The recovery of a Neoproterozoic palaeomagnetic record from South China permits the evaluation of palaeogeographic models for this tectonic block to be evaluated. Whilst consistent with some geological data, a proposition that the block was sited between eastern Australia and western Laurentia fails to satisfy the palaeomagnetic data. Instead, a location adjacent to western Australia is indicated by correlation of Late Neoproterozoic to Cambrian palaeomagnetic data from the two blocks. Contiguity of ~1000 Ma mobile belts suggests that this configuration had been attained by Early Neoproterozoic times whilst biogeographic data indicate that this essential location was retained into later Palaeozoic times. It appears that development of Neoproterozoic Rodinia reconstruction without recourse to palaeomagnetic data has produced some major inconsistencies which will require evaluation. In particular, we note that there is no correlation between palaeomagnetic poles from Laurentia and Australia during the critical 730–540 Ma period required by the geological evidence accommodated by the Rodinia reconstruction.

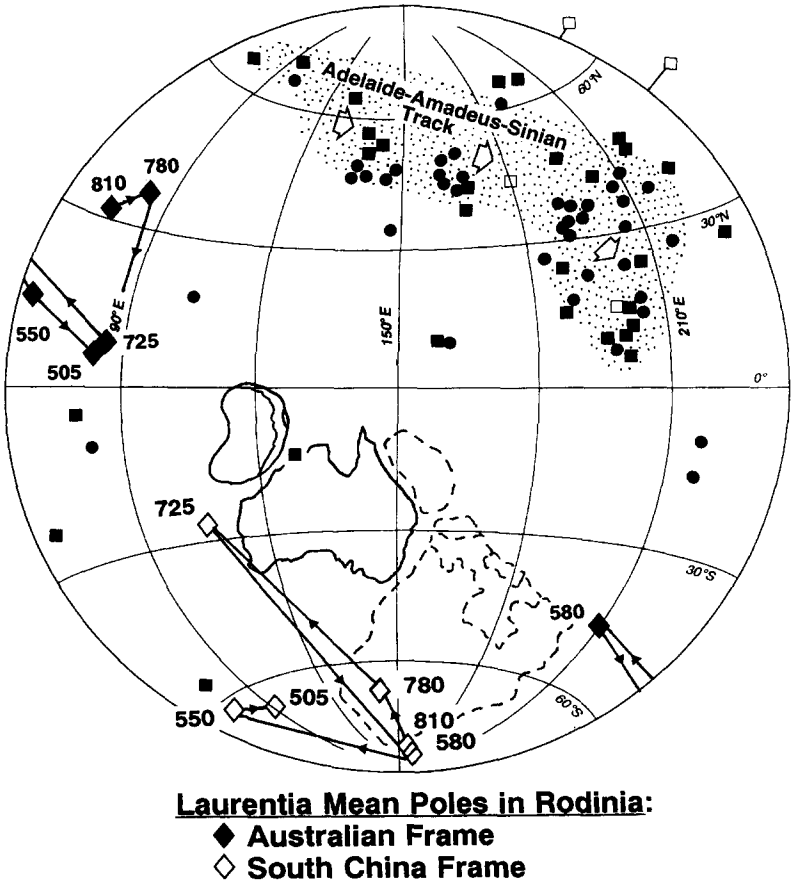


Fig. 15. Mean Neoproterozoic palaeomagnetic poles from the Laurentian Shield of North America after Powell et al. (1993) and Meert et al. (1995) rotated into the Australian reference frame according to the Rodinia reconstruction of Hoffman (1991), Dalziel (1991) and Moores (1991) and shown as closed diamonds. The open diamonds are the same poles rotated into the South China reference frame according to the reconstruction of Li et al. (1995). Note the absence of correlation with the Neoproterozoic APW track from Australia and South China. The dashed outlines show the locations of Laurentia and South China according to the Rodinia configuration.

Acknowledgments

This study is the outcome of a cooperative link between the Chinese Academy of Sciences and the Royal Society of London and was supported by the Royal Society, the Chinese Academy, the Chinese Natural Sciences Foundation and the University of Liverpool. Mr Shi Huai Ren of the Malong County Government and Mr Xiong Jia Yong, general engineer of the Geological Survey of Yunnan Province provided much useful support to our studies in the Malong region. Mr Liu Zhong Zun and his colleagues at the Kaiyang Phosphorous Mine kindly provided guidance and support in the Kaiyang district. Mr Chen Yu Lin of the Geological Survey Department of Guizhou Province provided us with reports to guide the fieldwork in Guizhou. Dr T.C. Rolph generously provided of his time and advice with the laboratory study. We are also grateful to Dr P.L. McFadden for providing programs to perform fold and reversal tests. We thank Dr Xixi Zhao and his colleagues for providing a preprint of their paper reviewing palaeomagnetic results from China and Professor Zhang Huimin for provided us with insight into the Chinese Neoproterozoic database. We would

also like to express our appreciation to Dr M.J. de Wit and Dr J. Meert for their criticisms of the manuscript and many suggestions for improvement.

References

Bell, R.T., Jefferson, C.W., 1987. An hypothesis for an Australian–Canadian connection in the late Proterozoic and the birth of the Pacific ocean. In: Pacific Rim Congress '87. Australian Institute of Mining and Metallurgy, pp. 39–50

Bond, G.C., Nickeson, P.A., Kominz, M.A., 1984. Breakup of a supercontinent between 625 Ma and 555 Ma: new evidence and implications for continental histories. Earth Planet. Sci. Lett. 70, 325–345.

Burrett, C.F., Richardson, R., 1980. Trilobite biogeography and Cambrian tectonic models. Tectonophysics 63, 155–192.

Burrett, C.F., Long, J., Stait, B., 1990. Early–Middle Palaeozoic biogeography of Asian terranes derived from Gondwana. In: McKerrow, W.S., Scotese, C.R. (Eds.), Memoir 12. Palaeozoic Palaeogeography and Biogeography Geological Society, London, pp. 97–104

Chen, X., Rong, J.Y., Rowley, D.E., Zhang, J., Zhang, Y.D., Zan, R.B., 1995. Is the early Palaeozoic Banxi Ocean in South China necessary? Geol. Rev. 41, 389–399.(in Chinese with English abstract)

Compston, W., Williams, I.S., Kirschvink, J.L., Zhang, Z., Ma, G., 1992. Zircon U–Pb ages for the Early Cambrian time scale. J. Geol. Soc. Lond. 149, 171–184.

Dalziel, I.W.D., 1991. Pacific margins of Laurentia and East Antactica–Australia as a conjugate rift pair: Evidence and implications for an Eocambrian supercontinent. Geology 19, 598–601.

Dalziel, I.W.D., Knoll, A., Moores, E., 1994. Late Precambrian Tectonics and the dawn of the Phanerozoic. GSA Today 4, 1–3.

Eisbacher, G.H., 1985. Late Proterozoic rifting, glacial sedimentation and sedimentary cycles in the light of Windermere deposition, Western Canada.. Palaeogeogr. Palaeoclimatol. Palaeoecol. 51, 231–254.

Eyre, J., 1994. Magnetic mineralogy of Chinese loess. Ph.D thesis, University of Liverpool.

Fairchild, I.J., Bradby, L., Spiro, B., 1994. Reactive carbonate in glacial systems: a preliminary synthesis of its creation, dissolution and reincarnation. In: Deynoux, M., Miller, J.M.G., Domack, E.W., Eyles, N., Fairchild, I.J., Young, G.M. (Eds.), Earth's Glacial Record. Cambridge University Press, Cambridge, pp. 176–192.

Fang, D.J., Van der Voo, R., Liang, Q.Z., 1988. Reconnaissance magnetostratigraphy of the Precambrian–Cambrian boundary section at Meishucun, southwest China. Cuadernos de Geologia Iberica 12, 205–222.

Frakes, L.A., 1979. Climates Through Geological Time. Elsevier, Amsterdam.

Gan, X.C., Zhao, F.Q., Li, H.M., Tang, X.X., Huang, J.Z., 1993. Single zircon U–Pb age of Banxi Group in Hunan. In: Isotopic Specialisations Committee of the geological Society of China (Eds.), The Evolution of Crust and Mantle, Diagenesis and Mineralisation Isotopic Geochemistry. Seismology Press, Beijing., pp. 10–11 (in Chinese with English abstract).

Harland, W.B., 1983. The Proterozoic glacial record. In: Medaris, L.G., Byers, C.W., Mickleson, D.M., Shanks, W.C. (Eds.), Proterozoic Geology. Geological Society of America Memoirs 161, 279–288

Hambrey, M.J., Harland, W.B., 1981. Earth's Pre-Pleistocene Glacial Record. Cambridge University Press, Cambridge.

Hoffman, P., 1991. Did the breakout of Laurentia turn Gondwanaland inside out? Science 252, 1409–1412.

Hu, S.L., Liu, H.Y., Lao, Q.Y., 1991. New geochronology of the Sinian System. Sci. Geol. Sinica 4, 325–336.

Jefferson, C.W., 1980. Correlation of Middle and Upper Proterozoic strata between northern Canada and south and central Australia, International Geology Congress 26, Resumes 2 (1980) 594.

Kirschvink, J.L., 1992. A palaeogeographic model for Vendian and Cambrian Time. In: Schopf, J.W., Klein, C. (Eds.), The Proterozoic Biosphere: A multidisciplinary study. Cambridge University Press, Cambridge, pp. 569–581.

Klootwijk, C.T., 1980. Early Palaeozoic palaeomagnetism in Australia. Tectonophysics 64, 249–332.

Knoll, A.H., Walter, M.R., 1992. Latest Proterozoic stratigraphy and Earth history. Nature 356, 673–678.

Kroner, A., 1977. Precambrian mobile belts of southern and western Africa — ancient sutures or sites of ensialic mobility? A case for crust evolution towards plate tectonics. Tectonophysics 40, 101–135.

Larson, E.E., Walker, T.R., 1982. A rock magnetic study of the Lower Massive Sandstone, Moenkopi Formation (Triassic) Gray Mountain area, Arizona. J. Geophys. Res. 87, 4819–4836.

Li, P., Liu, C., 1979. A palaeomagnetic study of the Sinian rocks along the east gorges area of the Yangzi River. Acta Geophys. Sinica 22, 281–288.

Li, Y., Li, Y., Sharps, R., McWilliams, M.O., Gao, Z., 1991. Sinian palaeomagnetic results from the Tarim block, western China. Precamb. Res. 49, 61–71.

Li, Z.X., Zhang, L., Powell, C. McA., 1995. South China in Rodinia; part of the missing link between Australia–East Antarctica and Laurentia? Geology 23, 407–410.

Lin, J.L., Fuller, M., 1990. Palaeomagnetism, North China and South China collision and the Tan–Lu Fault. Phil. Trans. R. Soc., Lond., Ser. A. 331, 589–598.

Lin, J.L., Fuller, M., Zhang, W.Y., 1985. Palaeogeography of the North and South China blocks during the Cambrian. J. Geodynamics 2, 91–114.

Liu, C., Liu, H.S., 1965. Several palaeomagnetic results from the Sinian in China. Geol. Sci. Sinica 1, 77–79.(in Chinese)

Liu, H.Y., 1991. The Sinian System in China, Science Press, Beijing.

Liu, H.Y., Zhang, Q.R., 1991. The Sinian System in China. In:

Tu, G.Z. (Ed.), Advances in Science of China — Earth Sciences, vol. 3, 1989–1990. Science Press, Beijing, pp. 1–24.

Liu, C., Liang, Q., Li, Y., 1988. Magnetostratigraphy of the Precambrian–Cambrian boundary section at Meishucun, Jinning, Yunnan. Zhongguo Kexue (B) 3, 314–320.(in Chinese)

Liu, Y., Yang, W., Hayao, M., Yasushisa, A., Takafumi, Y., Katsumi, Y., 1993. Palaeogeographic history of the North China continental block, Qinling geological block and Yangtze continental block since the Palaeozoic era. Geol. Sci. Tech. Inf. 12, 17–21.(in Chinese with English abstract)

Long, J., Burrett, C., 1989. Fish from the Upper Devonian of the Shan–Thai terrane indicate proximity of east Gondwana and South China terranes. Geology 17, 811–813.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. Geophys. Res. Lett. 17, 159–162.

Ma, G.G., Li, H.Q., Xie, W.X., 1980. A discussion on the isotopic ages of Sinian in western gorges and the Sinian time scale in China. Bull. Yichang Inst. Geol. Miner. Res., CAGS 1 (1), 39–55.(in Chinese with English abstract)

McCabe, C., Elmore, R.D., 1989. The occurence and origin of Late Palaeozoic remagnetisation in the sedimentary rocks of North America. Rev. Geophys. 27, 471–494.

McElhinny, M.W., Giddings, J.W., Embleton, B.J.J., 1974. Palaeomagnetic results and late Precambrian glaciations. Nature 248, 557–561.

McFadden, P.L., 1990. A new fold test for palaeomagnetic studies. Geophys. J. Int. 103, 163–169.

McKerrow, W.S., Scotese, C.R., Brasier, M.D., 1992. Early Cambrian continental reconstructions. J. Geol. Soc. Lond. 149, 599–606.

McMenamin, M.A.S., Schulte McMenamin, D.L., 1990. The Emergence of Animals; the Cambrian Breakthrough. Columbia University Press, New York.

Meert, J.G., Van der Voo, R., 1994. The Neoproterozoic (1000–540 Ma) glacial intervals: No more snowball earth? Earth Planet. Sci. Lett. 123, 1–13.

Meert, J.G., Van der Voo, R., Ayub, S., 1995. Palaeomagnetic investigation of the Neoproterozoic Gagwe lavas and Mbozi Complex, Tanzania and the assembly of Gondwana. Precamb. Res. 74, 244–255.

Moores, E.M., 1991. Southwest US–East Antarctic (SWEAT) connection: an hypothesis. Geology 19, 425–428.

Opdyke, N.D., Huang, K., Xu, G., Zhang, W.Y., Kent, D.V., 1987. Palaeomagnetic results from the Silurian of the Yangtze paraplatform. Tectonophysics 139, 123–132.

Park, J.K., 1992. Did Laurentia pass over the south pole during earliest Cambrian time? Current Research, Part E. Geological Survey of Canada, paper 92-1E, pp. 11–22.

Piper, J.D.A., 1982. The Precambrian palaeomagnetic record: the case for the Proterozoic Supercontinent. Earth Planet. Sci. Lett. 59, 61–89.

Piper, J.D.A., 1987. Palaeomagnetism and the Continental Crust. Wiley, New York.

Piper, J.D.A., Zhang. Q.R., 1996. Palaeomagnetism of Neoproterozoic glacial rocks of the Huabei Shield: the North China Block in Gondwana. Tectonophysics (in press).

Powell, C.M., McElhinny, M.W., Li, Z.X., Meert, J.G., Park, J.K., 1993. Palaeomagnetic constraints on timing of the Neoproterozoic breakup of Rodinia and Cambrian formation of Gondwana. Geology 21, 889–892.

Robertson, D.J., France, D.E., 1994. Discrimination of remanence — carrying minerals in mixtures using isothermal remanent magnetisation acquisition curves. Phys. Earth Planet. Int. 82, 223–234.

Schemerhorn, L.J.G., 1974. Late Precambrian mixtites: Glacial and/or non-glacial? Am. J. Sci. 274, 673–824.

Schmidt, P.W., Williams, G.E., 1995. The Proterozoic climate paradox: Equatorial palaeolatitude for Marinoan glaciation near sea level in South Australia. Earth Planet. Sci. Lett. 134, 107–124.

Setiabudidaya, D., Piper, J.D.A., Shaw, J., 1994. Palaeomagnetism of the (Early Devonian) Lower Old Red Sandstones of South Wales: implications to Variscan overprinting and differential regional rotations. Tectonophysics 231, 257–280.

Tang, J.F., 1993. A report on the field excursion on the Banxi Group in Hunan Province in 1990. J. East China Geol. Inst. 16, 420–424.(in Chinese)

Tauxe, I., Badgley, C., 1984. Transition stratigraphy and the problem of remanence lock-in times in the Siwalik red beds. Geophys. Res. Lett. 11, 611–613.

Wang, Z., Van der Voo, R., 1993. Pervasive remagnetisation of Palaeozoic rocks acquired at the time of Mesozoic folding in the South China Block. J. Geophys. Res. 98, 1729–1741.

Wang, Z., Van der Voo, R., Wang, Y., 1993. A new Mesozoic apparent polar wander loop for South China: palaeomagnetism of Middle Triassic rocks from Guizhou Province. Earth Planet. Sci. Lett. 115, 1–12.

Wang, Z., Van der Voo, R., Wang, Y., 1995. Palaeomagnetic results from Late Sinian/Early Cambrian rocks in Guizhou Province, South China Block. In: Subbarao, K.V. (Ed.), Magnetism, Rocks to Superconductors. Geological Society of India, Memoir 29, 159–182.

Williams, G.E., 1993. History of the Earth's obliquity. Earth Sci. Rev. 34, 1–45.

Williams, G.E., Schmidt, P.W., Embleton, B.J.J., 1995. Comment on 'The Neoproterozoic (1000-540 Ma) glacial intervals: No more snowball Earth?' by J.G.Meert and R. Van der Voo. Earth Planet. Sci. Lett. 131, 115–122.

Yang, Z., Cheng, Y., Wang, H., 1986. The Geology of China. Clarendon Press, Oxford.

Young, G.M., 1995. Are Neoproterozoic glacial deposits preserved on the margins of Laurentia related to the fragmentation of two supercontinents? Geology 23, 153–156.

Zhang, H.M., Zhang, W.Z., Li, P., 1983. Palaeomagnetism of the Sinian System of Eastern Yangzi Gorges in China. Bull. Tianjin Inst. Miner. Res. 6, 57–68.(in Chinese)

Zhang, H.M., Zhang, W.Z., Elston, D.P., 1991. Palaeomagnetic study of Middle and Late Proterozoic rocks in Jixian, North China, Acta Geophys. Sinica 34, 602–615 (in Chinese) and Chin. J. Geophys. 34, 631–647 (in English).

Zhang, H.M., Zhang, W.Z., Elston, D.P., 1993. Palaeomagnetism of Middle and Upper Proterozoic rocks in Jixian, North China and in Jinshan, Hubei Province, South China. International Exchange on Earth Sciences (XIV). Selected Papers from International Scientific and Technological Projects (1986–1990). Geological Publishing House, Beijing, pp. 198–214 (in Chinese).

Zhao, X., Coe, R.S., Gilder, S.A., Frost, G.M., 1996. Palaeomagnetic constraints on the palaeogeography of China: implications for Gondwanaland. Aust. J. Earth Sci. 43, 643–672.

Zhou, X., Hu., S., Ren, S., Li, J., 1993. Isochronological constraints to multistage tectonic evolution of the continental crust in southeastern China. In: Li, J. (Ed.), Geophysical Structure and Geological Evolution of Continental Lithosphere in southeastern China. Metallurgical Industry Press, Beijing, pp. 69–66 (in Chinese).

Zhu, H., He, Z.X., 1987. A palaeomagnetic and tectonic study of Palaeozoic strata from the north margin of Alashan terrane (in Chinese). In: Biostratigraphy and Tectonics of Palaeozoic strata along the Alashan terrane margin. Wuhan Geological College Press, Wuhan, pp. 154–159.